INTRODUCTION TO SPEECH SIGNAL PROCESSING

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Slides were prepared with the help of Metilda N J, Ph.D. Scholar

BROAD OUTLINE OF THE COURSE

BASICS

- Speech
- DSP
- Statistics

ASR - Statistical/Conventional

- HMM-GMM
- Hybrid DNN-HMM
- TDNN and Chain Models
- Language Models

ASR - Deep Learning

- E2E - ASR
- RNNs/LSTMs
- Transformers
- Language Models

SELF-SUPERVISED MODELS

- BERT
- GPT
- Mockingjay
- APC
- Wav2vec 2.0
- HuBERT

SPEAKER VERIFICATION

- x-vectors
- r-vectors
- s-vectors
- ECAPA-TDNN

TTS

- USS
- HMM_GMM
- WaveNet
- E2E TTS
- Tachotron
- Fastspeech
- Fastspeech-2

TENTATIVE EVALUATION SCHEME

weekly assignments: need min to be able to appear for Final..

Two Mid-sem evaluations – 25% each

Final evaluation - 40% each

Mid-Sems and Final will be done “physically” at TCS centres



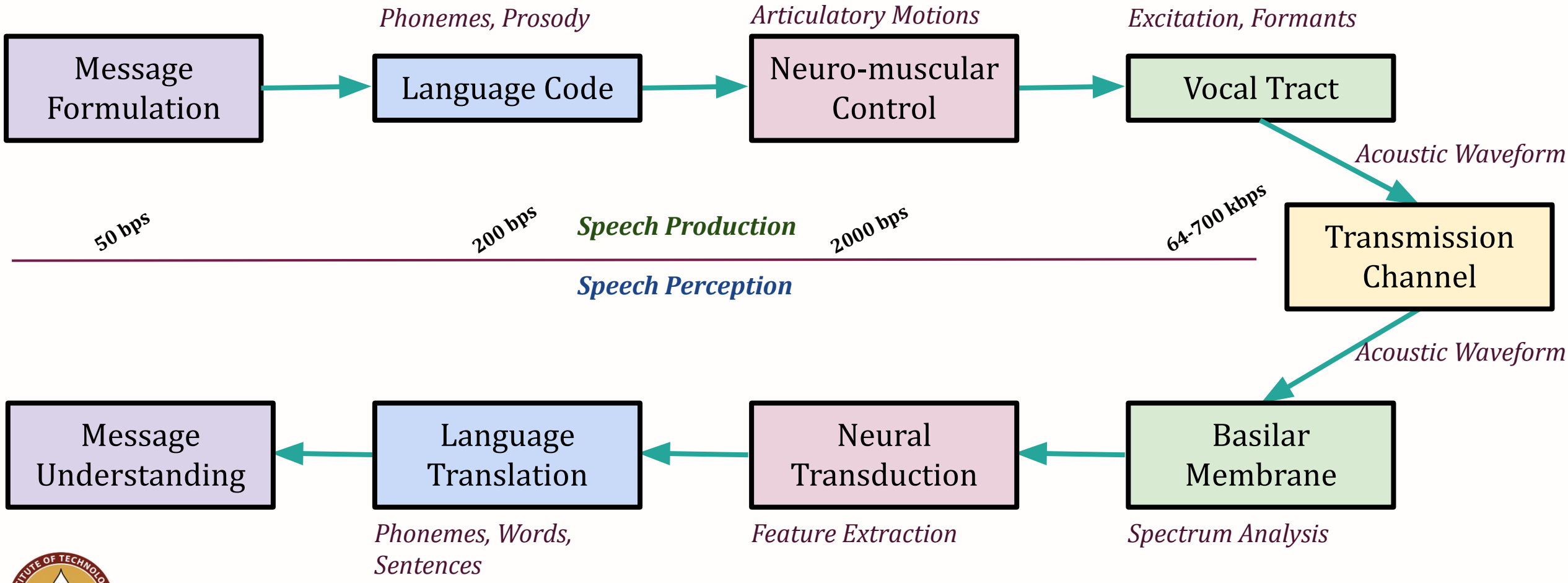
PRIMARY PURPOSE OF SPEECH - Communication

- **Message:** You want to ask your friend whether he/she wants to join for Lunch

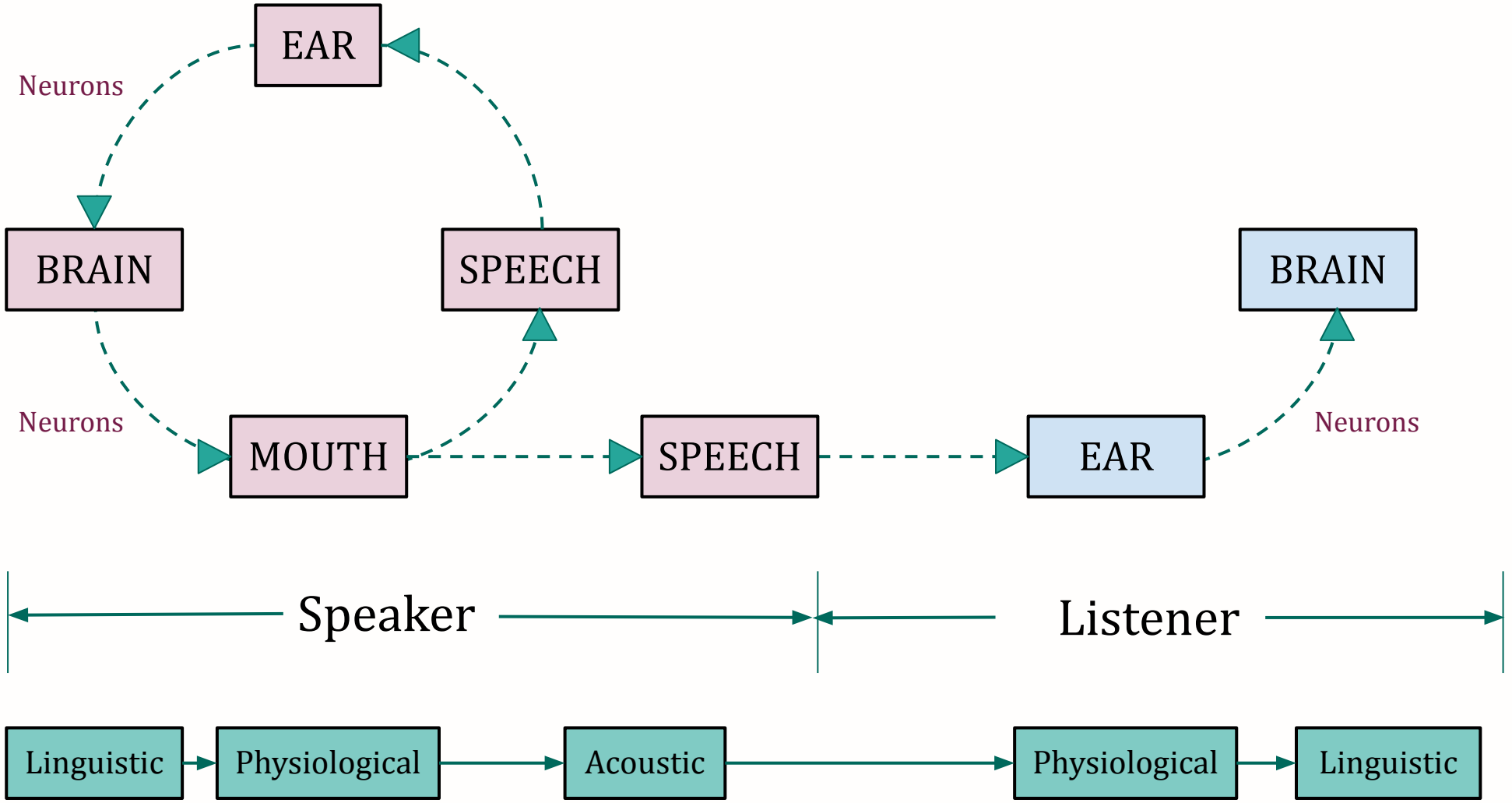
- You can frame this questions in multiple ways and in multiple languages
  - *Do you want to go for Lunch*
  - *Want to join me for lunch?*
  - *Lunch? khaane ke liye ja raha hoon or sapada polama?*

- **Output speech:** Once you formulate the sentence and language, you move your mouth articulators appropriately (controlled by brain) and the spoken audio comes out

SPEECH – To communicate message/information

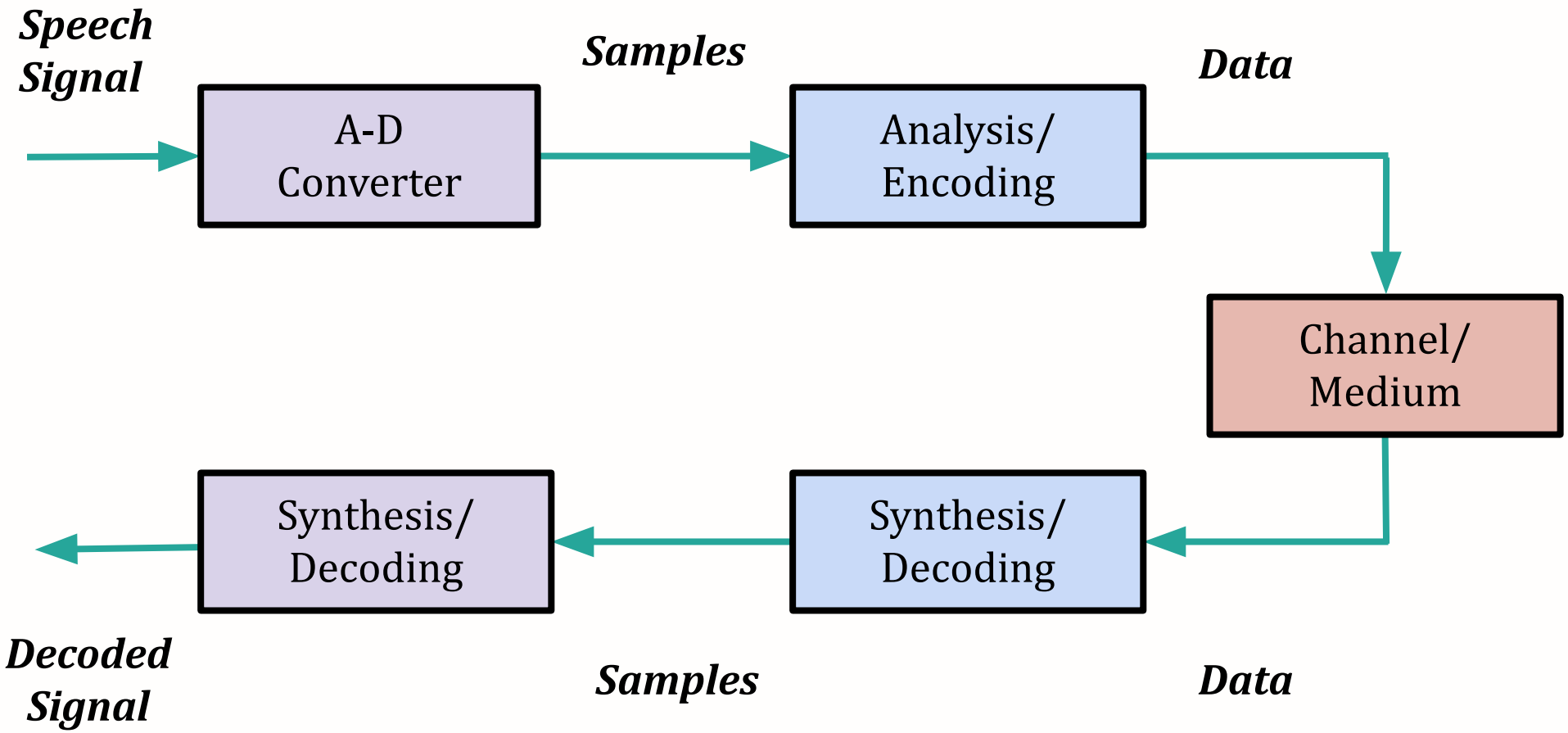


SPEECH CHAIN - LISTENER RECEIVES MESSAGE

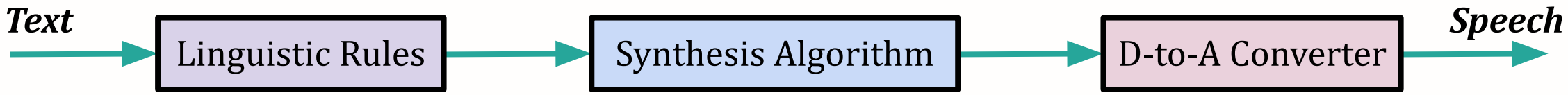


SPEECH CODING/COMPRESSION

Application I: How to store and transmit speech efficiently?



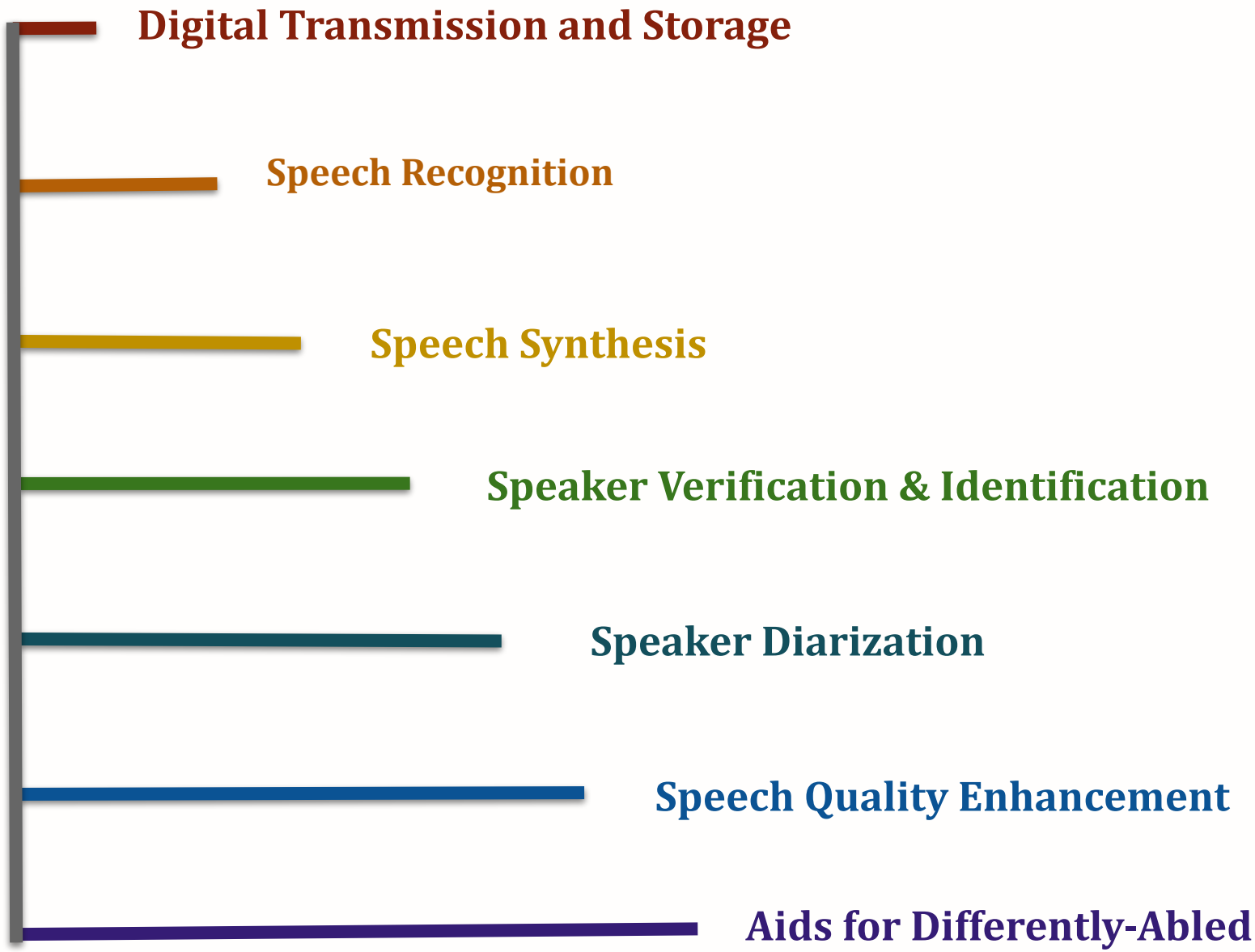
TEXT TO SPEECH (TTS) – Machine talking to Humans



PATTERN MATCHING – Humans talking to Machines



Digital Speech Processing Applications



DSP FUNDAMENTALS

A QUICK RECAP

- Sampling
- Fourier Series
- CT/DT Fourier Transform
- DFT
- z-Transforms



SAMPLING

- We work with **computers** —> so can handle **digital** signals only, i.e. numbers with finite precision
- A continuous-time (or analog) signal can be stored in a digital computer, in the form of equidistant **discrete points** or **samples**
- **Sampling Theorem:** Under certain conditions, we can reconstruct the original continuous-time signal from samples!
- What is the minimum sampling rate that will reconstruct the signal back properly?
  - **By Nyquist-Shannon Sampling Theorem**
  - **Sampling Frequency ≥ 2 x Maximum Frequency in the Signal**
- Incorrect Sampling Frequency – Loss of information (Aliasing) sampled signal look like a resultant of a slowly varying signal
- Discrete sampled signals are then **quantised** (finite precision) to get digital signals.

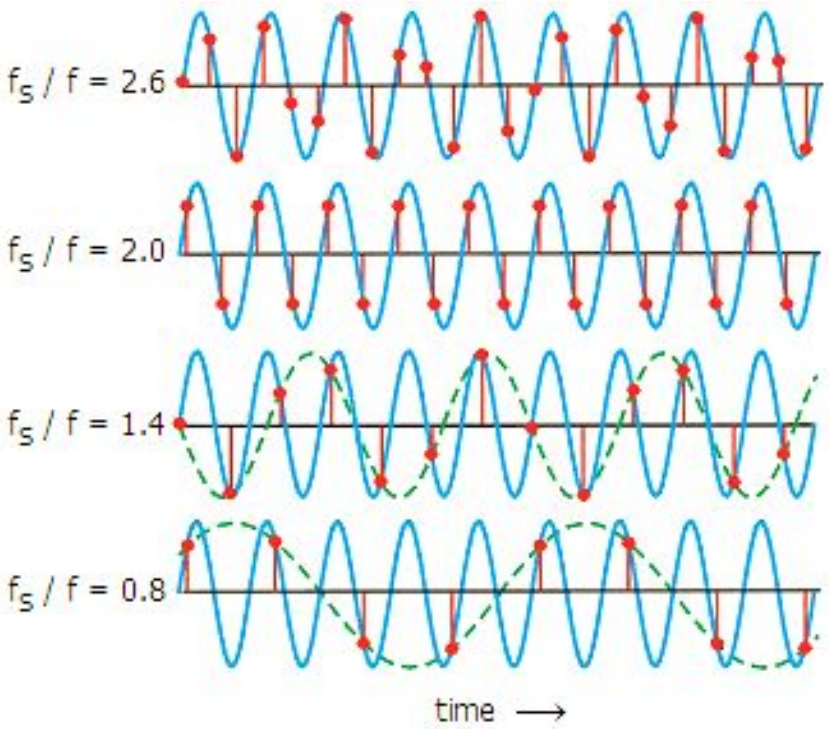


Fig: Different sampling rates and what they mean

http://195.134.76.37/applets/AppletNyquist/Appl_Nyquist2.html

SAMPLING & SPEECH SIGNALS

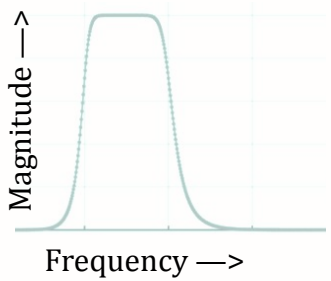
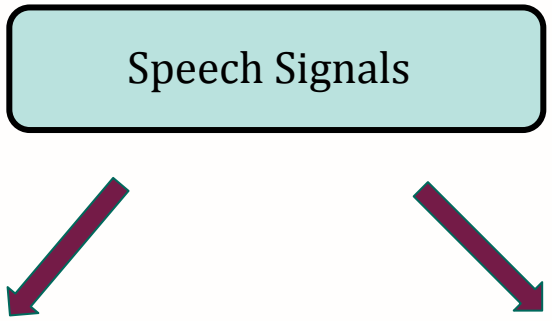


Fig: Freq. Resp. of Band Pass Filter



Narrow-Band

- ➔ **Telephone Speech**
- ➔ Telephone Channel acts Band Pass Filter (3**00 to 3.4 kHz**)
- ➔ By Sampling Theorem, **sampling frequency is 8 kHz**
- ➔ **8 k samples every second**

Wide-band

- ➔ **Current Standard**
- ➔ Also known as **HD-voice (50 - 7 kHz)**
- ➔ By Sampling Theorem, **sampling frequency is 16 kHz**
- ➔ **16 k samples every second**

https://www.analogictips.com/rf-microwave-bandpass-filter-implementations-part-1-distributed-filters/

SOME STANDARDS AND APPROACHES

Speech – Preferred 16 kHz sampling frequency and 16-bit PCM quantisation

➔ **Telephone** – A-law, mu-law standards are used (64 kbits/s)

➔ **Parametric approaches** – based on Code-Excited Linear Prediction (Mobile/Cellular)

$$s[n] = \sum_{k=1}^{P} \alpha_k \left(s[n-k]\right)$$

➔ **GSM and CDMA** – (7-13 kbits/s)

Audio/Music – 44.1 kHz and 48 kHz are quite popular

MP3 – are “lossy” and operate at lower bit-rates but are “perceptually” CD quality

ANALYSING FREQUENCIES & HUMANS

Seeing colours is essentially analysing frequencies

Different rods and cones react to different frequencies or colours

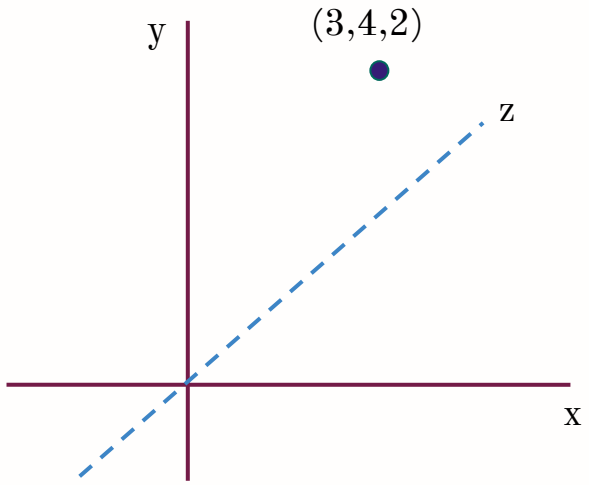
Similarly, hearing essentially involves analysis of frequencies!

More on this later!

EXPANSION IN TERMS OF BASIS FUNCTIONS

Recall: Any point in 3-D can be expressed as linear combination of unit vectors

$$\hat{i} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{j} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$



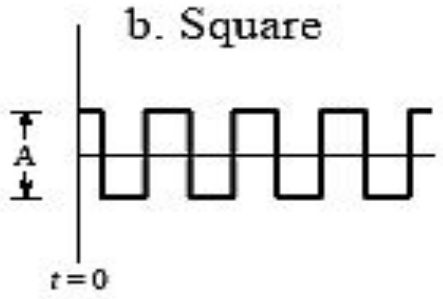
$$\begin{pmatrix} 3 \\ 4 \\ 2 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + 4 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 2 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

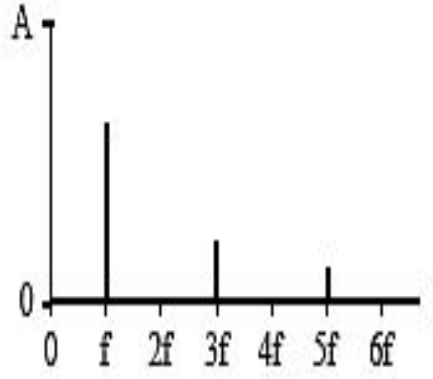
EXPANSION IN TERMS OF BASIS FUNCTIONS

Similarly, under reasonable conditions,

- A signal can be expressed as a linear combination of complex sinusoids of different frequencies – Fourier series/transform

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{inx}{L}}$$



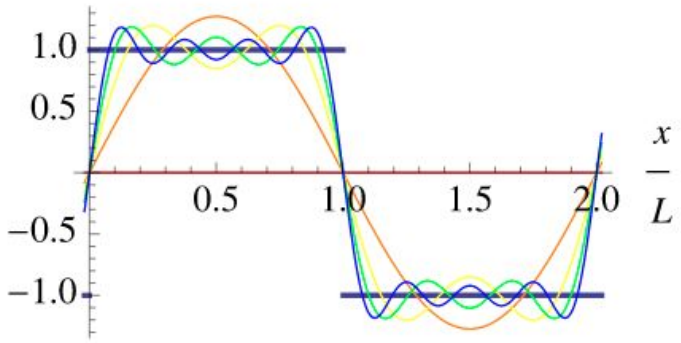


- Transforms help "transform" from time-variable space to frequency-variable space

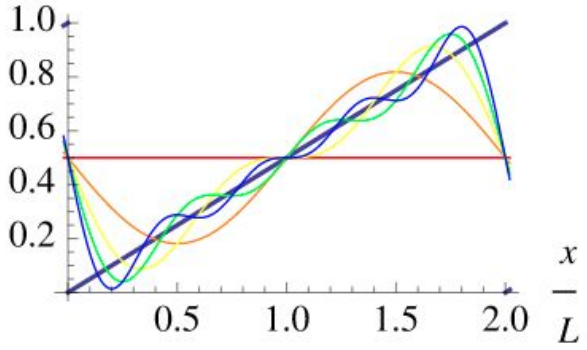
- It is two different views of the same signal

FOURIER SERIES

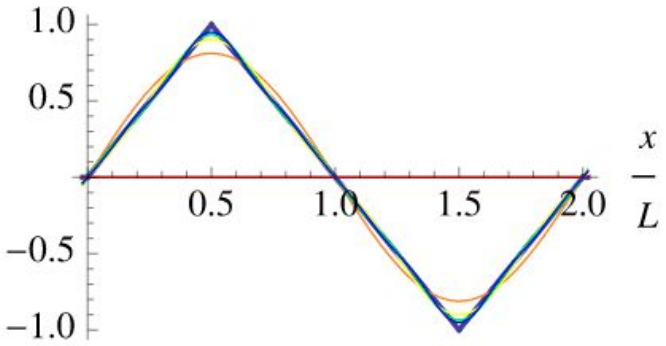
square wave



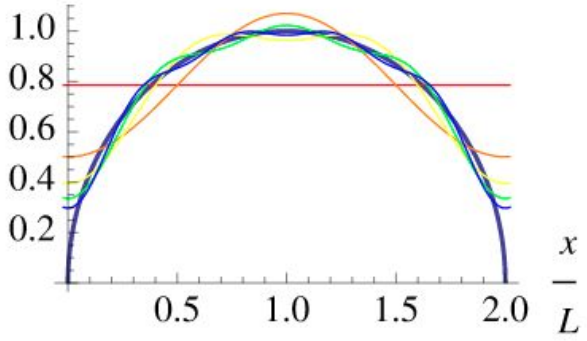
sawtooth wave



triangle wave



semicircle



- ➔ Any **periodic** signal can be represented as a sum of complex sinusoids with different frequencies
- ➔ Used in Analysis: find frequency components in a signal
- ➔ Both continuous and discrete time versions are available

https://mathworld.wolfram.com/FourierSeries.html

FOURIER SERIES

CTFS:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{\frac{inx}{L}}$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx, n = 0, \pm 1, \pm 2, \ldots$$

- ➔ Input — Continuous & Periodic
- ➔ Output — Discrete & Aperiodic

DTFS:

$$f([n]) = \sum_{k=0}^{N-1} c_k e^{i\omega_0 kn}$$

$$a^k = \frac{1}{N} \sum_{n=0}^{N-1} x(n) e^{\frac{-i2\pi kn}{N}}$$

- ➔ Input — Discrete & Periodic
- ➔ Output — Discrete & Periodic

DTFS leads to DFT which is usually the transform we use when analysing on computers

FOURIER TRANSFORM

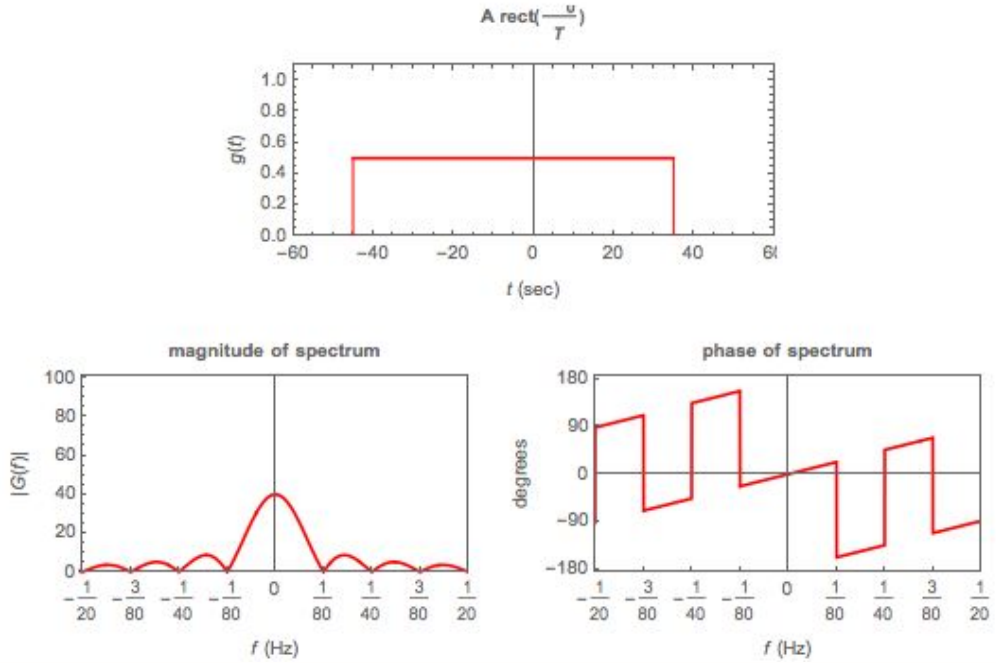


Fig: Box Function & its Fourier Transform

- ➔ Any **aperiodic** signal with finite energy can have Fourier Transform
- ➔ Used in Analysis of frequency components in a signal
- ➔ Both continuous and discrete time versions are available

https://demonstrations.wolfram.com/RectangularPulseAndItsFourierTransform/

FOURIER TRANSFORM

CTFT:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)\, e^{-i\omega t}\, dt$$

- ➔ Input — Continuous & Aperiodic
- ➔ Output — Continuous & Aperiodic

DTFT:

$$X(\omega) = \sum_{n=-\infty}^{\infty} x(n)\, e^{-i\omega n}$$

- ➔ Input — Discrete & Aperiodic
- ➔ Output — Continuous & Periodic

DISCRETE FOURIER TRANSFORM

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{\frac{-i2\pi kn}{N}}$$

- ➔ Input — Discrete & Periodic
- ➔ Output — Discrete & Periodic
- ➔ k = 0, 1, 2, ……..(N-1)

- ➔ Discrete in both domains
- ➔ Used in analysis of frequency components in a signal
- ➔ Used in actual digital signal processing with computers

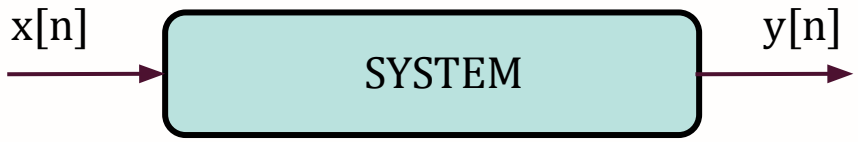
z-TRANSFORM

$$X(Z) = \sum_{-\infty}^{\infty} x(n)\, z^{-n}$$

- ➔ A System manipulates signals to give other signals
- ➔ z-transform is a tool for analysing the i**nteraction between signals and systems**
- ➔ For discrete signals and systems
- ➔ A signal without DTFT can still have Z-Transform



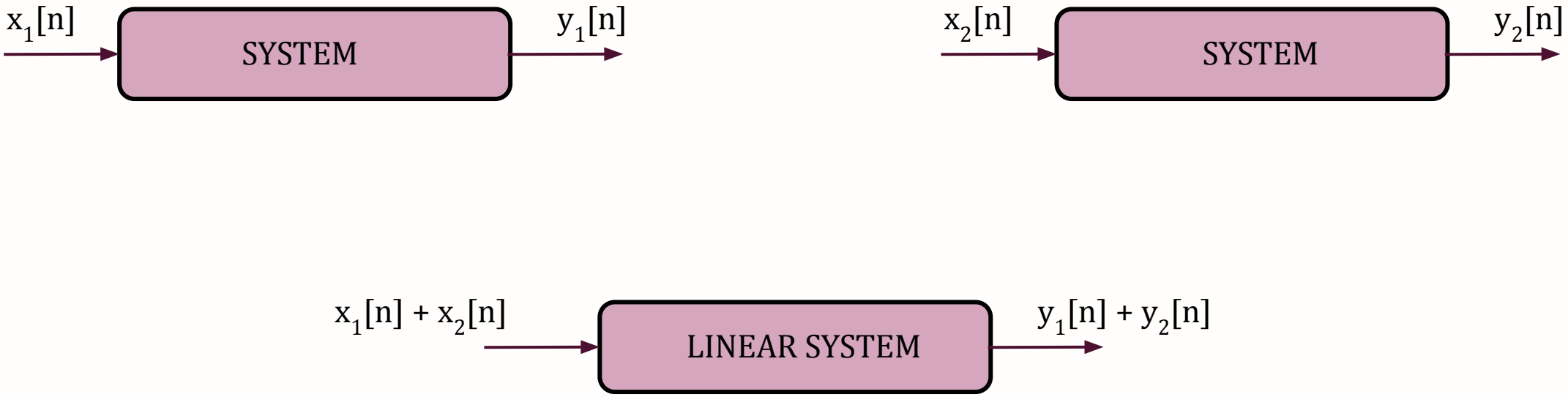
REVIEW OF SYSTEMS



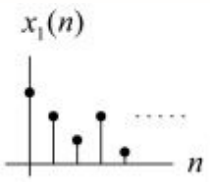
PROPERTIES OF SYSTEMS:

- ➔ Linear/Nonlinear
- ➔ Time variant/invariant
- ➔ Memoryless/Memory
- ➔ Causal/Anticausal/Non-causal
- ➔ Stable/Unstable

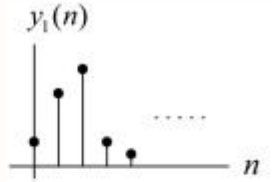
LINEAR/NONLINEAR

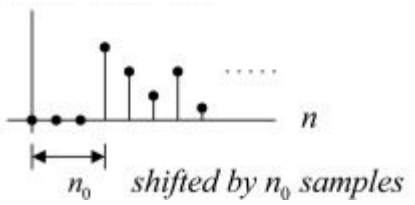


TIME VARIANT/INVARIANT

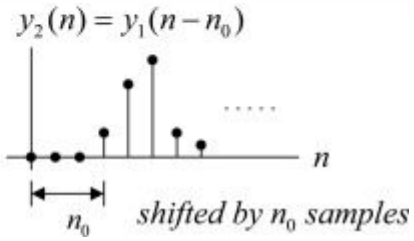


x[n] → SYSTEM → y[n]





x[n-k] → TIME INVARIANT SYSTEM → y_2[n-k]



MEMORYLESS/MEMORY

CAUSAL/ANTICAUSAL



Memoryless:

- ➔ y[n] depends on the current sample of x[n] only
- ➔ y[n]=2x[n]

Memory:

- ➔ y[n] depends on other samples too
- ➔ y[n]=x[n]+x[n-2]

Causal:

- ➔ y[n] depends on present and past samples of x[n]
- ➔ y[n]=x[n]+x[n-1]+x[n-2]

Anticausal:

- ➔ y[n] depends on present and future samples of x[n]
- ➔ y[n]=x[n]+x[n+2]

Noncausal:

- ➔ y[n] depends on present, past and future samples of x[n]
- ➔ y[n]=x[n-1]+x[n]+x[n+5]

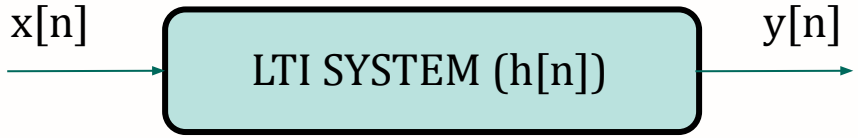
STABLE/UNSTABLE



Stable System:

- ➔ y[n] is bounded for bounded x[n] - BIBO Stability
- ➔ y[n] = 2x[n] - **Stable**
- ➔ y[n] = x[n]/n - **Unstable**

Linear & Time-Invariant (LTI) SYSTEMS



LTI SYSTEM:

y[n] = x[n] ★ h[n]

★ ⟶ Convolution

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

By Time invariance property:

$$\delta[n] \longrightarrow h[n]$$

$$\delta[n-k] \longrightarrow h[n-k]$$

By Linearity & Time invariance property:

$$a_1\delta[n-k_1] + a_2\delta[n-k_2] \longrightarrow a_1 h[n-k_1] + a_2 h[n-k_2]$$

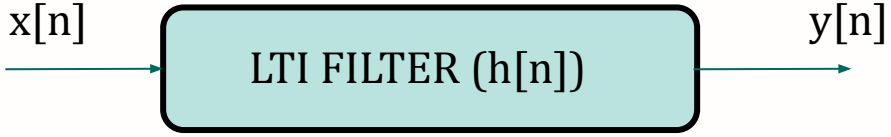
Any x[n] can be represented as:

$$\ldots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \ldots$$

Output of LTI System for given x[n]:

$$\ldots + x[-1]h[n+1] + x[0]h[n] + x[1]h[n-1] + \ldots$$

LTI SYSTEMS - FILTERS



LTI SYSTEM:

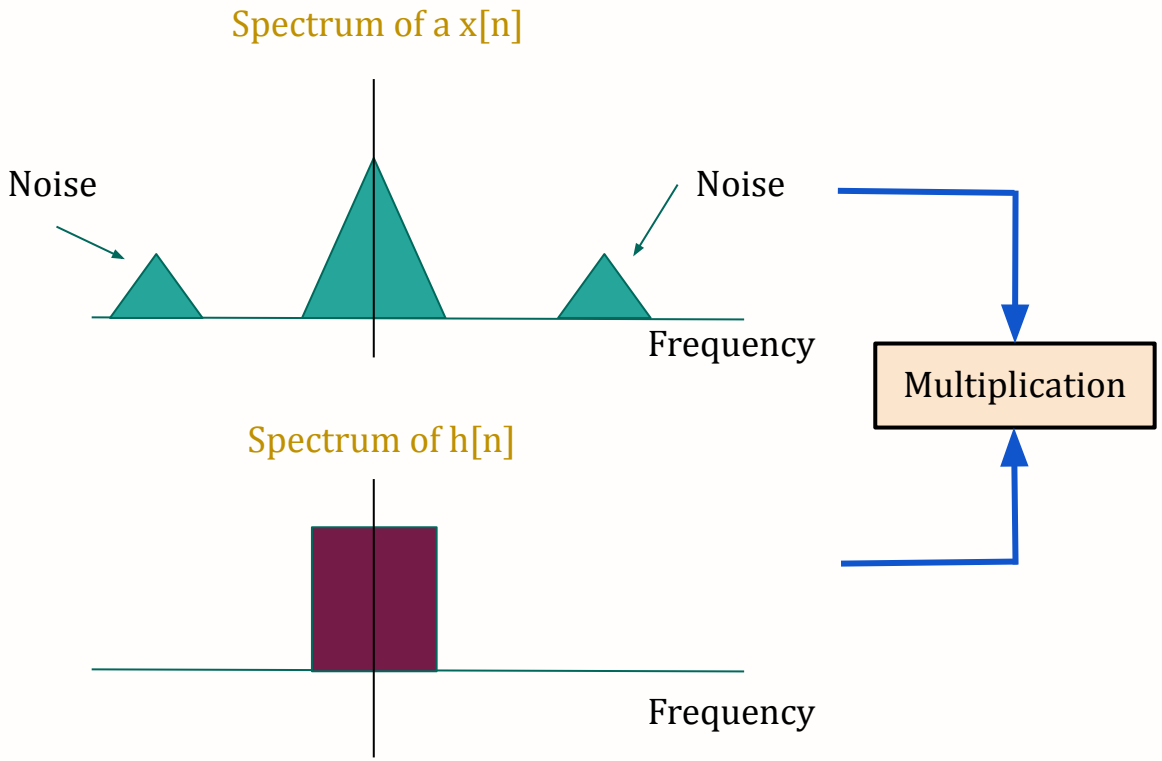
$y[n] = x[n] \bigstar h[n]$

$\bigstar$ ⟶ Convolution

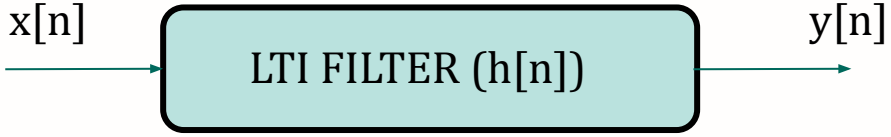
$$y\,[n] = \sum_{k=-\infty}^{\infty} x\,[k]\,h\,[n-k]$$

Convolution:

- ➔ **Time domain** - **Convolution**
- ➔ **Frequency Domain** - **Multiplication**



LTI SYSTEMS - Filters



LTI SYSTEM:

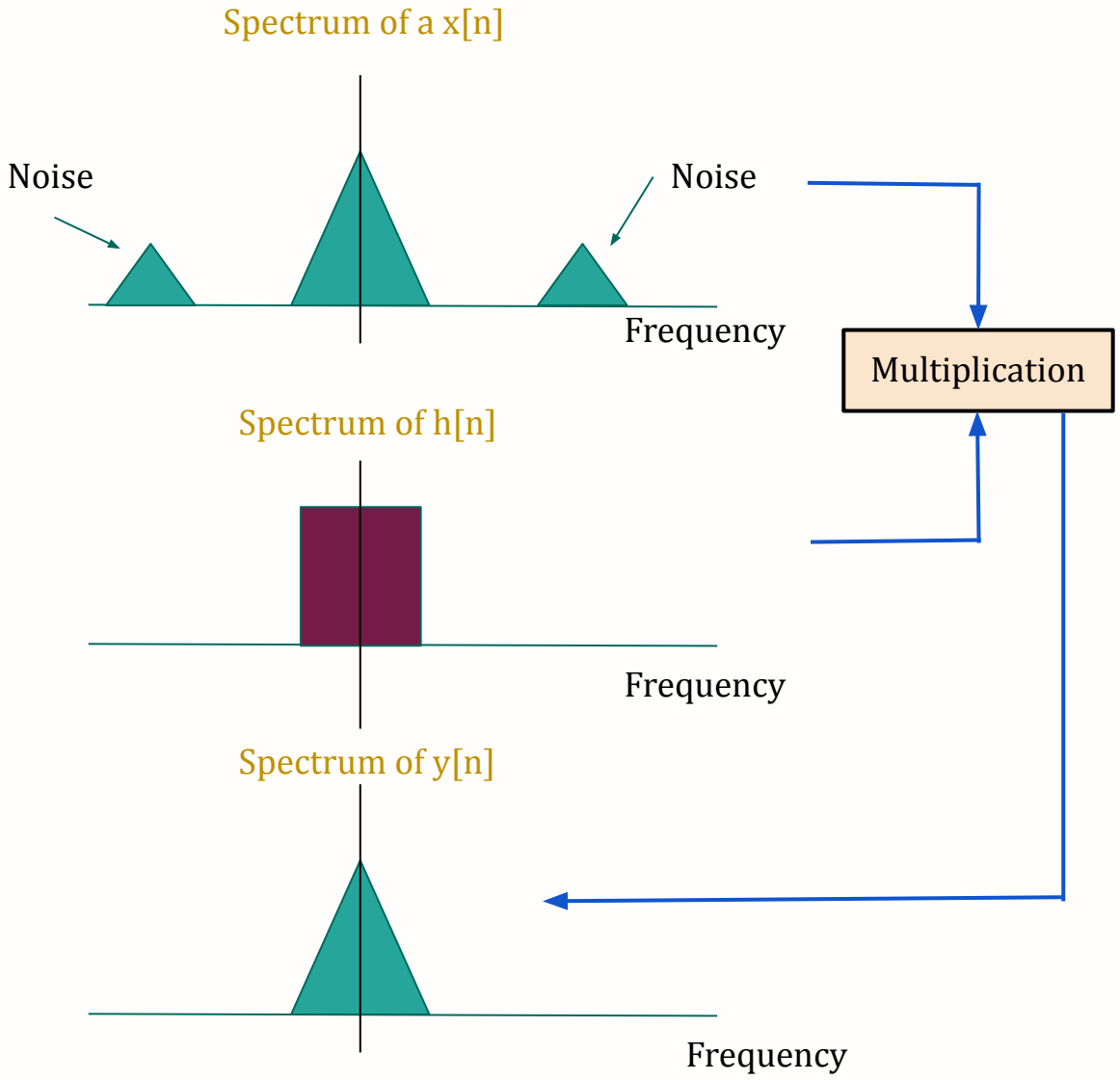
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$\bigstar$ ⟶ Convolution

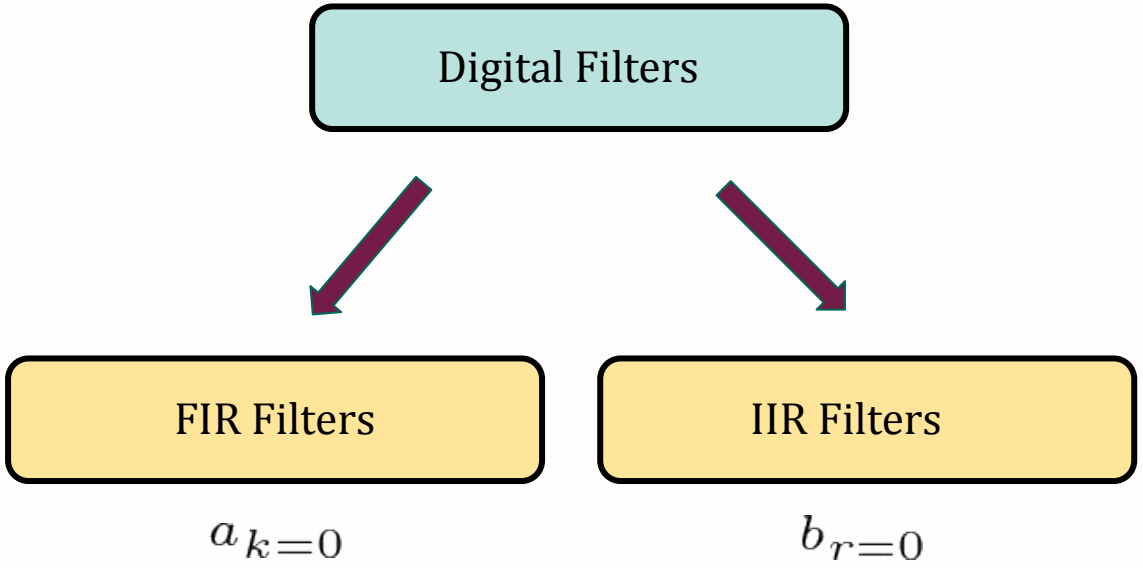
$$y[n] = \sum_{k=-\infty}^{\infty} x[k]\, h[n-k]$$

Convolution:

- ➔ **Time domain** - **Convolution**
- ➔ **Frequency Domain** - **Multiplication**



RATIONAL FILTERS – An important Class of Systems



- Does not take past outputs into account
- Takes current and past inputs (causal)

- Takes past inputs and outputs into account (causal)

- Filter Difference Equation

$$y[n] - \sum_{k=1}^{N} a_k y[n-k] = \sum_{r=1}^{M} b_r x[n-r]$$

- Filter Transfer Function

$$H(Z) = \frac{Y(Z)}{X(Z)} = \frac{\sum_{r=0}^{M} b_r z^{-r}}{1 - \sum_{k=1}^{N} a_k z^{-k}}$$

DIGITAL FILTERS

➔ Filter Transfer Function

$$H(Z) = \frac{Y(Z)}{X(Z)} = \frac{\sum_{r=0}^{M} b_r z^{-r}}{1 - \sum_{k=1}^{N} a_k z^{-k}}$$

$$H(Z) = K \frac{(s - z_1)(s - z_2)}{(s - p_1)(s - p_2)(s - p_3)}$$

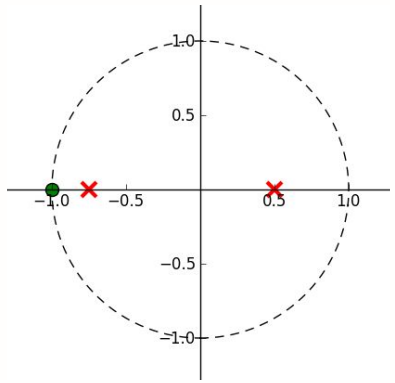


Fig: Example pole-zero plot

➔ Roots of the numerator polynomial are called the **zeros** of the filter

➔ Roots of the denominator polynomial are called the **poles** of the filter

➔ For a filter to be **stable**, the ROC must include unit circle

➔ For the filter to be **causal**, ROC must be outside of the outermost pole

➔ For a filter to be **stable and causal**, all the poles must be within unit circle

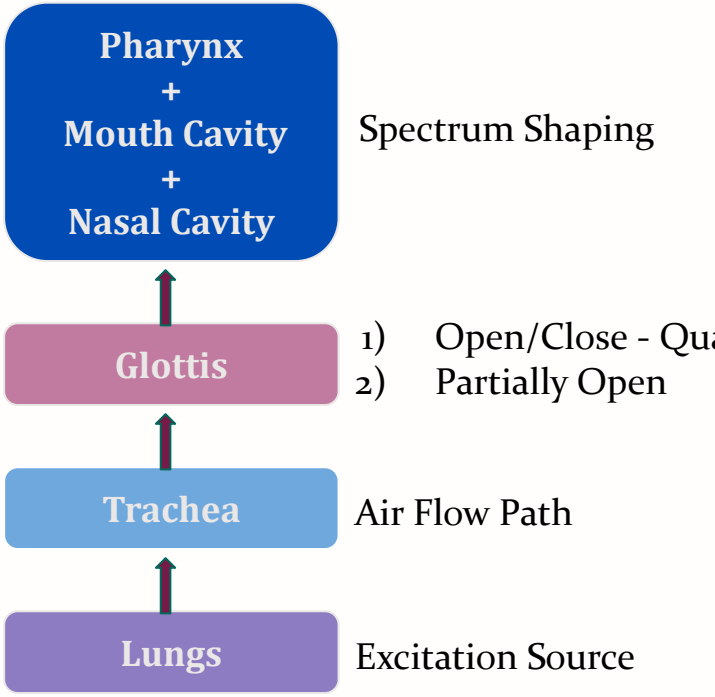
https://www.dsprelated.com/showcode/244.php

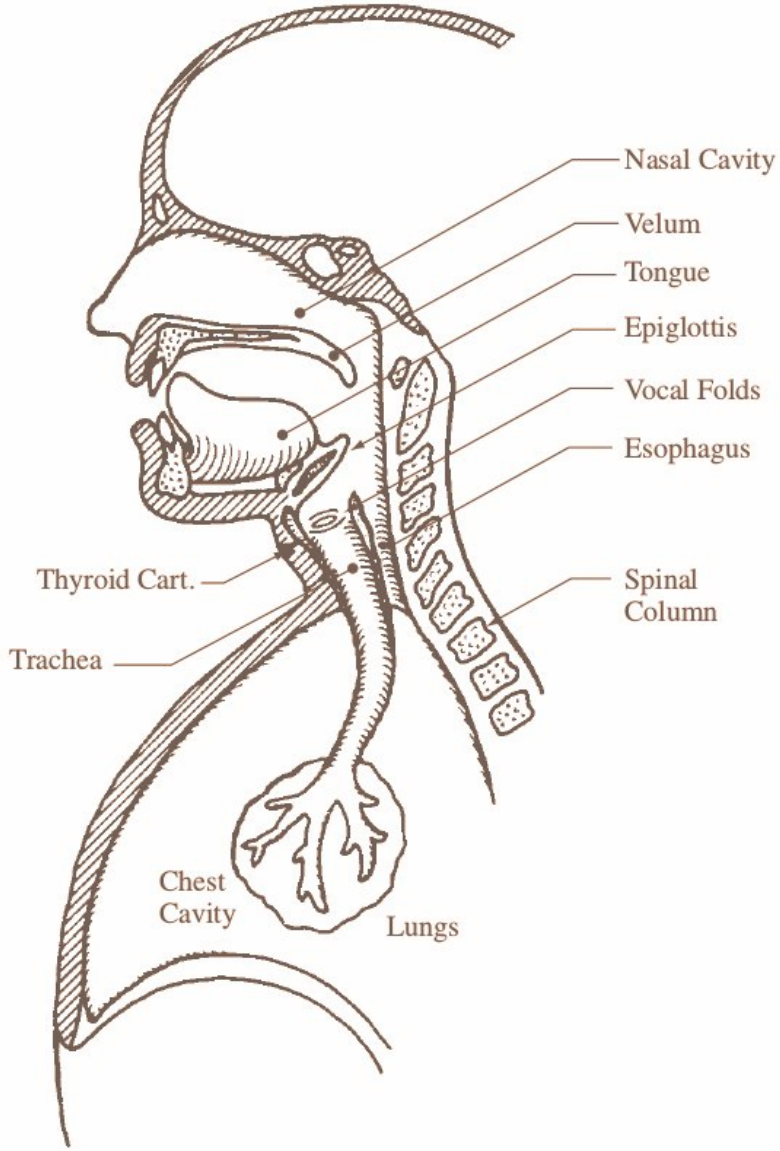
SPEECH PRODUCTION

- Human Vocal Tract
- Voiced and Unvoiced Speech
- Phonemes
- Voice-Filter Model
- Pitch
- Speech & STFT



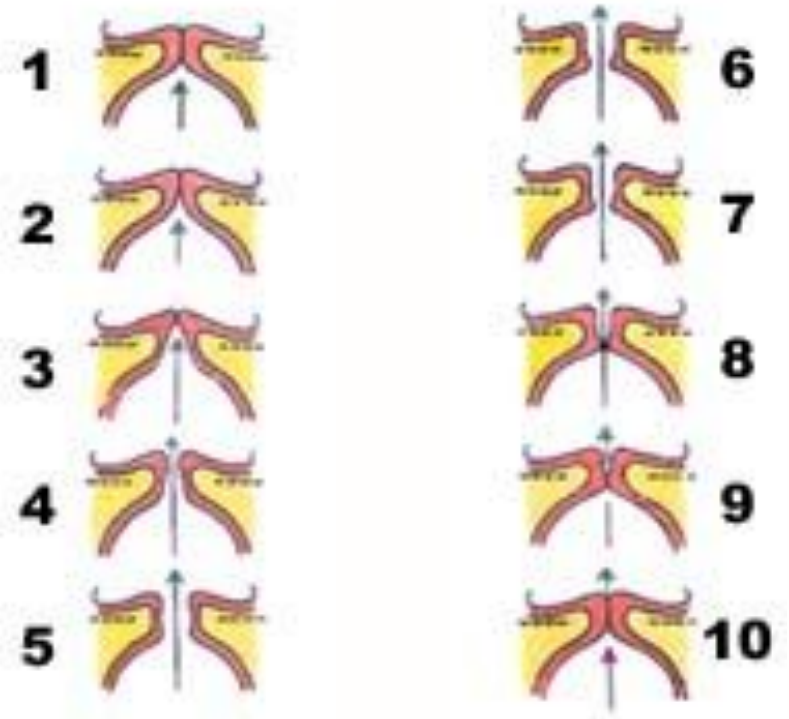
HUMAN VOCAL TRACT





Ref: Theory and Applications of Digital Speech Processing, 1st edition

HUMAN VOCAL CORD - VIBRATION CYCLE

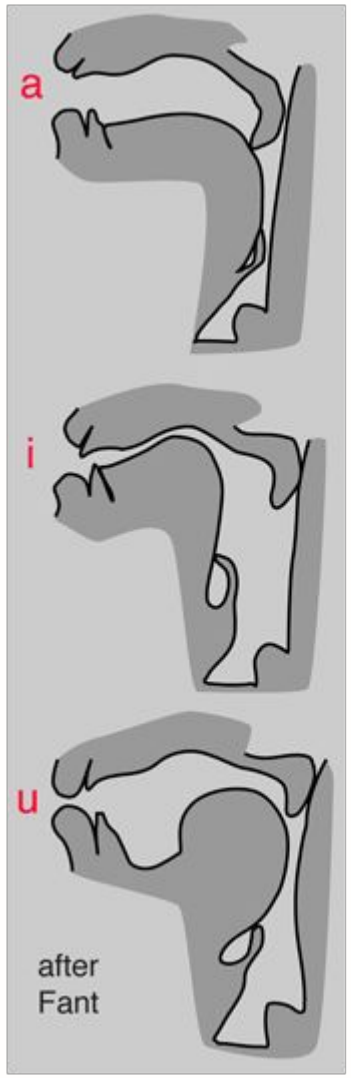


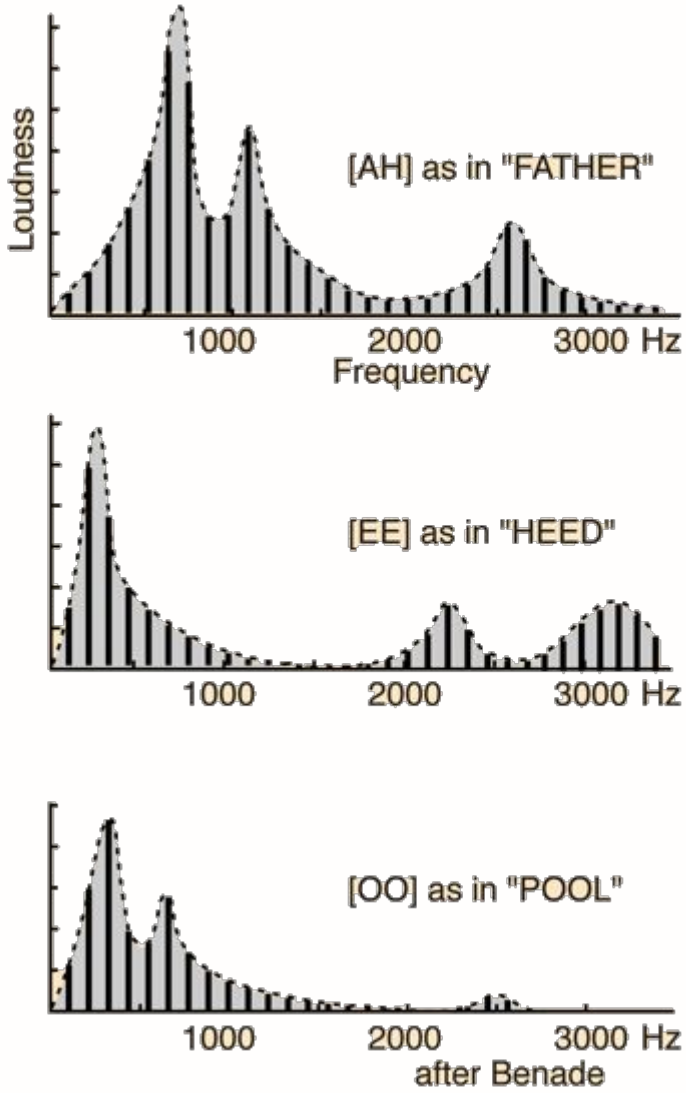
➔ Time period of the vibration cycle depends on the height & weight

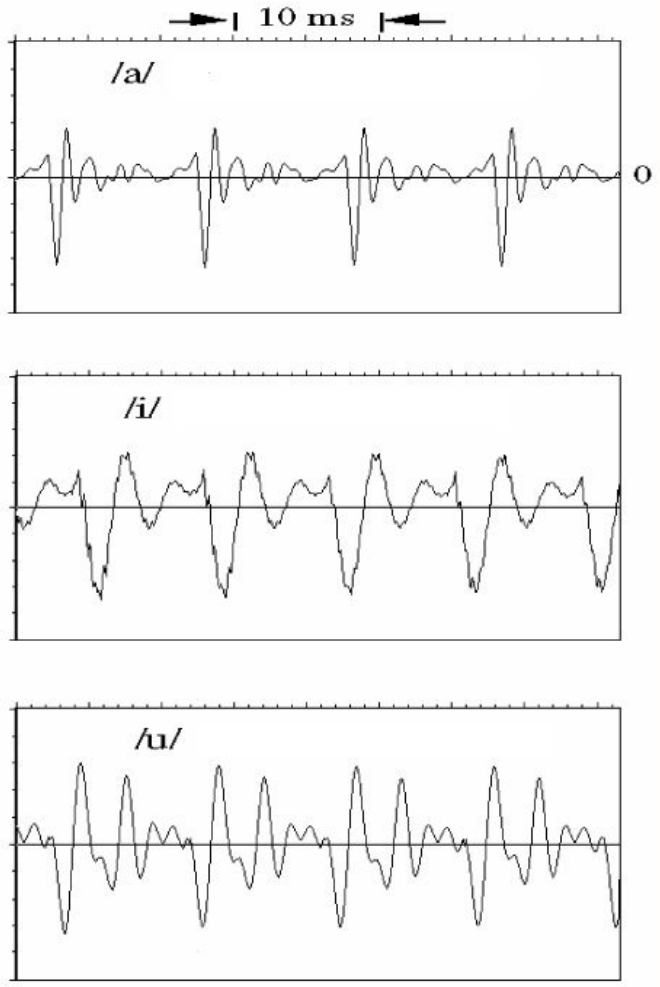
➔ $\text{Time period}_{\text{Men}} > \text{Time period}_{\text{Women}} > \text{Time period}_{\text{Children}}$

https://voicefoundation.org/health-science/voice-disorders/anatomy-physiology-of-voice-production/understanding-voice-production/

MOUTH CONFIGURATIONS & SOME VOWELS







http://hyperphysics.phy-astr.gsu.edu/hbase/Music/vowel.html ,
http://research.spa.aalto.fi/publications/theses/lemmetty_mst/chap3.html

VOWELS & DIPHTHONGS

All Spoken Vowels are “voiced” :

- ➔ about 10-12 different vowels
- ➔ /a/, /i/, /u/

In English:

- ➔ “written” vowels - “a”, “e”, “i”, “o” ,“u”

In Hindi:

- ➔ spoken and written closely follow each other

PHONEMES

Phonemes are like alphabets for speech

Phonemes are the basic units from which speech is formed

Written Form:

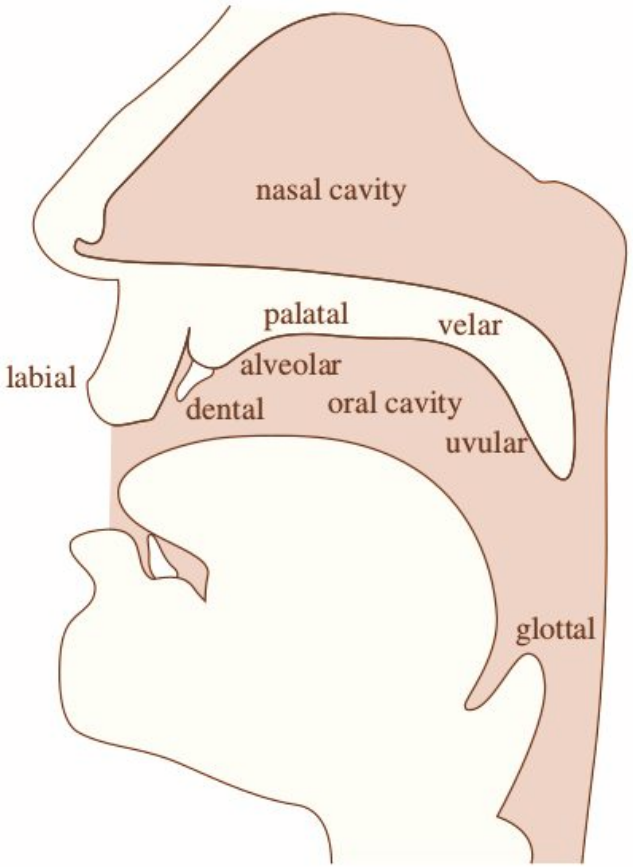
- ➔ English - 26 alphabets
- ➔ Hindi - 44/52 alphabets

PHONEME SET FOR ENGLISH

IPA Phoneme	ARPAbet	Example	IPA Phoneme	ARPAbet	Example
/i/	IY	beet	/ŋ/	NX	sing
/I/	IH	bit	/p/	P	pat
/ɚ/	AXR	butter	/t/	T	ten
/ɛ/, /e/	EH	bet	/k/	K	kit
/æ/	AE	bat	/b/	B	bet
/a/	AA	Bob	/d/	D	debt
/ʌ/	AH	but	/g/	G	get
/ɔ/	AO	bought	/h/	HH	hat
/o/	OW	boat	/f/	F	fat
/ʊ/	UH	book	/θ/	TH	thing
/u/	UW	boot	/s/	S	sat
/ə/	AX	about	/ʃ/, /sh/, /š/	SH	shut
/ɨ/	IX	roses	/v/	V	vat
/ɝ/	ER	bird	/ð/	DH	that
/e^y/	EY	bait	/z/	Z	zoo
/a^w/	AW	down	/ʒ/, /zh/, /ž/	ZH	azure
/a^y/	AY	buy	/tʃ/, /č/	CH	church
/$ɔ^y$/	OY	boy	/dʒ/, /j/	JH	judge
/y/	Y	you	/ʍ/	WH	which
/w/	W	wit	/l/	EL	battle
/r/	R	rent	/m/	EM	bottom
/l/	L	let	/n/	EN	button
/m/	M	met	/ɾ/	DX	batter
/n/	N	net	/ʔ/	Q	(glottal stop)

Ref: Theory and Applications of Digital Speech Processing, 1st edition

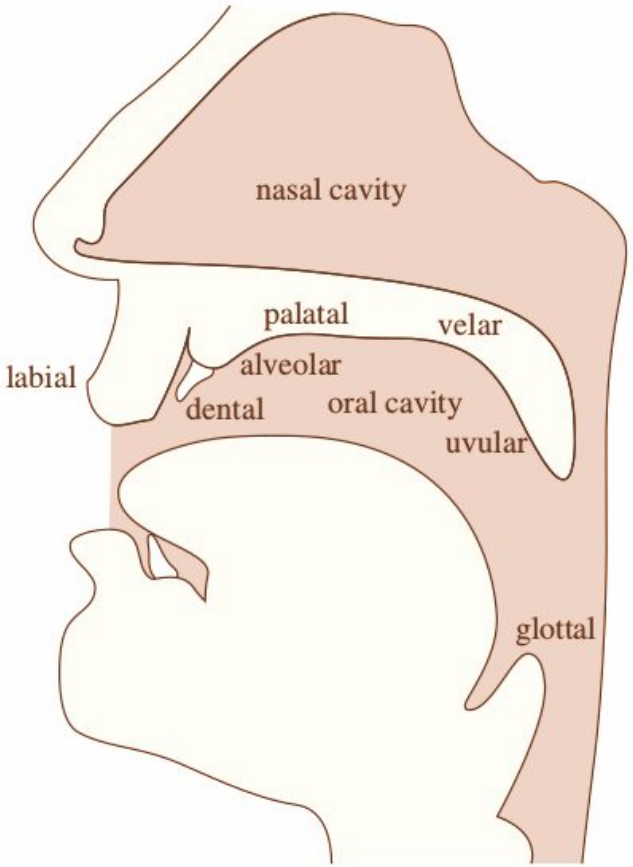
SOUNDS-ARTICULATION MANNER



- ➔ Glide /w/, /l/, /r/, /y/
- ➔ Nasal /m/, /n/
- ➔ Stop /p/, /t/, /k/, /b/, /d/, /g/
- ➔ Fricative /f/, /z/, /h/, /v/
- ➔ Voicing /b/, /d/, /g/, /v/, /w/, /l/ ,r/, /y/
- ➔ Mixed source /j/
- ➔ Whispered /h/

Ref: Theory and Applications of Digital Speech Processing, 1st edition

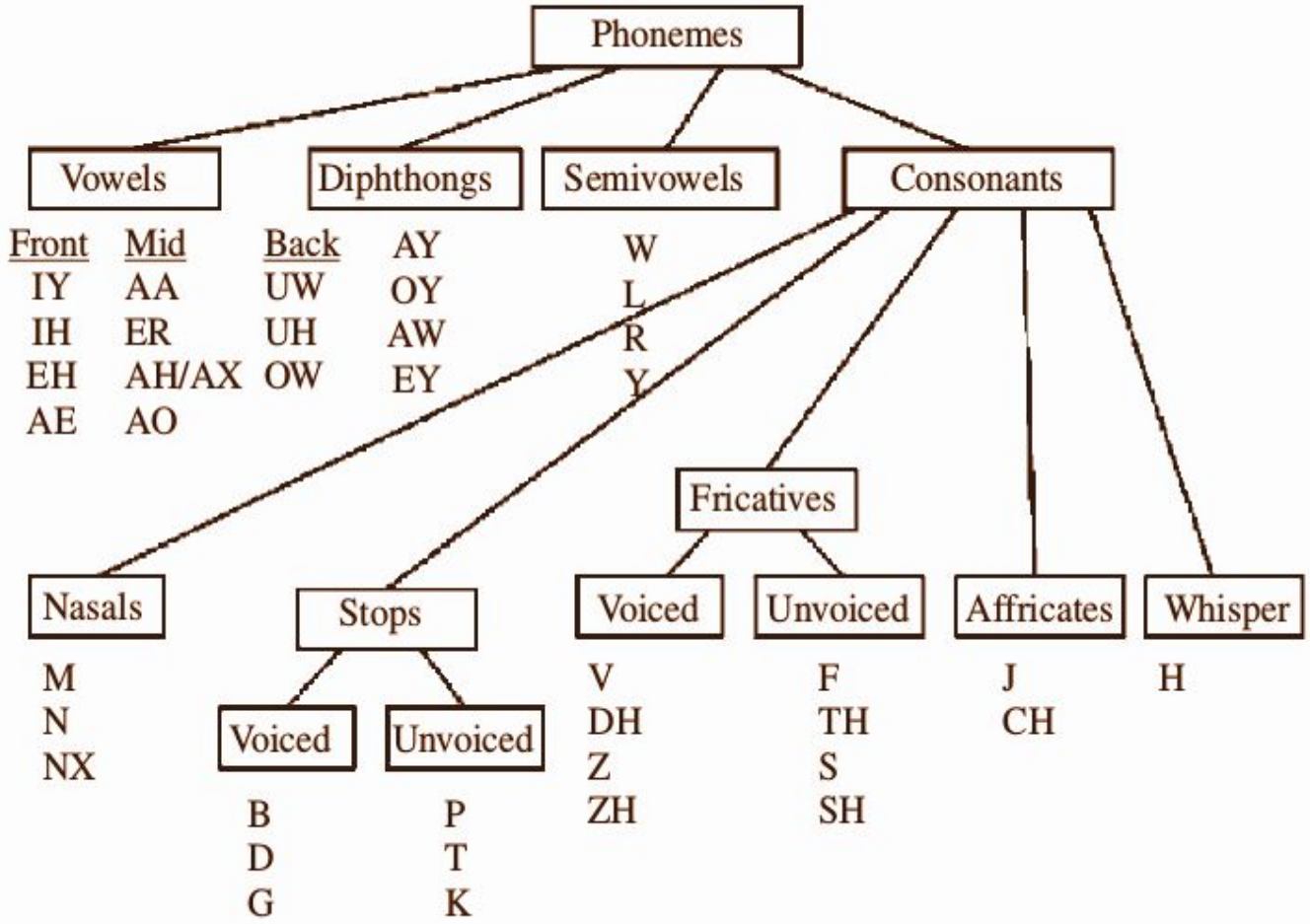
SOUNDS-ARTICULATION PLACE



- ➔ Labial /p/, /b/,/m/,/w/
- ➔ Labiodental /f/, /v/
- ➔ Dental /Θ/
- ➔ Alveolar /t/, /d/, /s/, /z/, /n/, /l/
- ➔ Palatal /r/
- ➔ Velar /k/, /g/
- ➔ Pharyngeal /h/

Ref: Theory and Applications of Digital Speech Processing, 1st edition

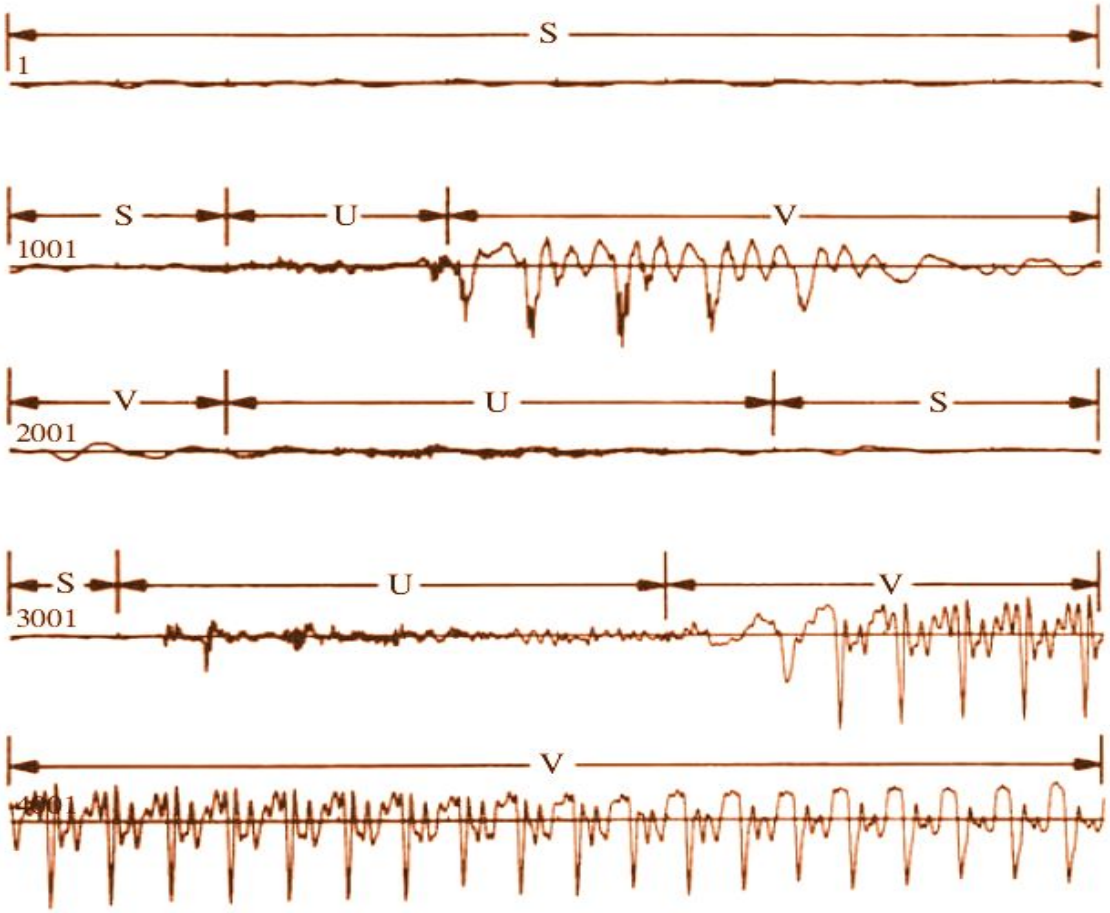
PHONEME SET FOR ENGLISH



- **Phone** is the basic unit of speech
- Every language has 40-60 distinct sounds
- Phones of closer languages may overlap
- These distinct sound units or phonemes are produced by appropriately positioning jaw, tongue, lips etc
- **Phoneme** is the "canonical" form of phone
- Every language has its own set of Phonemes

Ref: Theory and Applications of Digital Speech Processing, 1st edition

DIFFERENT SPEECH SOUNDS & WAVEFORM



S - Silence
U - Unvoiced Speech
V - Voiced Speech

➔ There is no clear cut boundary between silence and speech

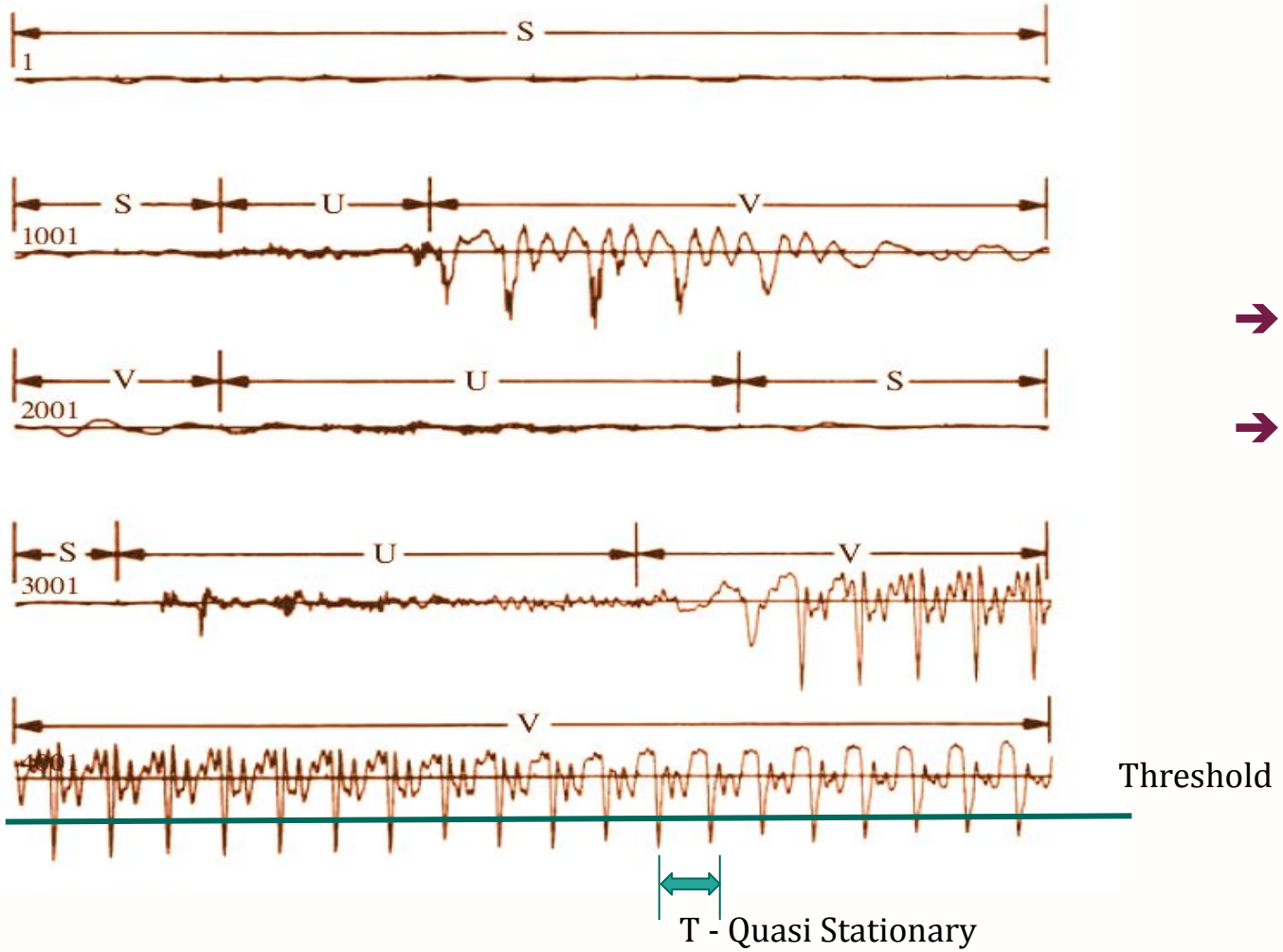
➔ There is no clear cut boundary between voiced and unvoiced

➔ **How to identify gender?**

➔ **How to identify Adult/Child Speech?**

Ref: Theory and Applications of Digital Speech Processing, 1st edition

DIFFERENT SPEECH SOUNDS & WAVEFORM



- **How to identify gender?**
- **How to identify Adult/Child Speech?**
  - Find 1/T
  - Then decide

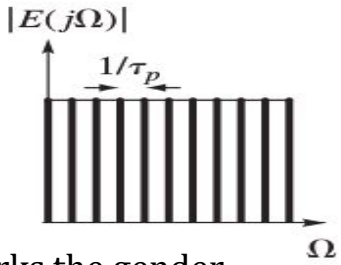
Ref: Theory and Applications of Digital Speech Processing, 1st edition

SPEECH PRODUCTION MODEL

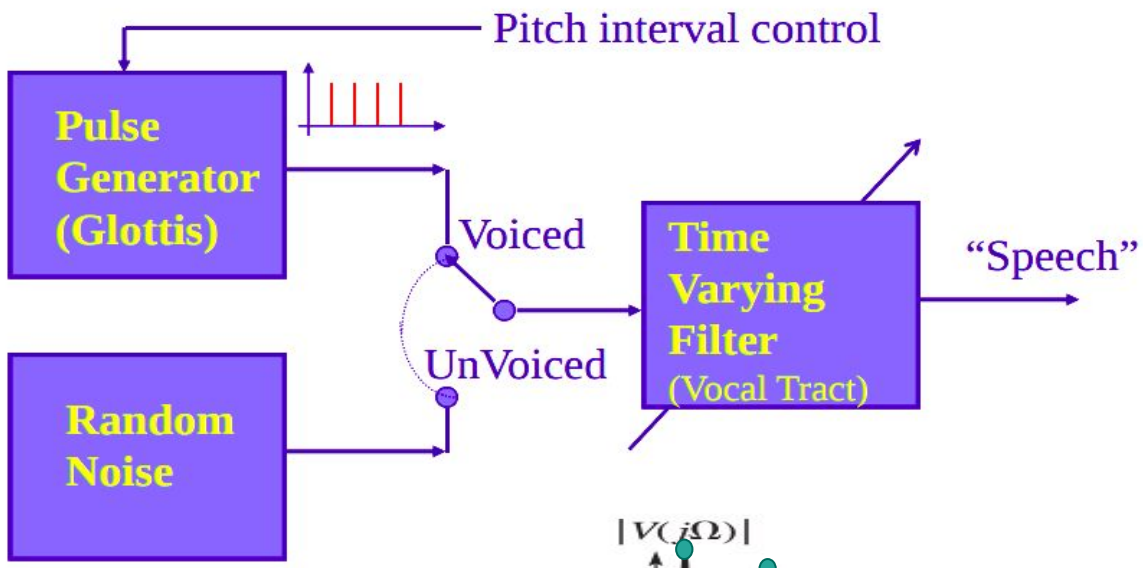
➔ It is the mathematical equivalent of speech

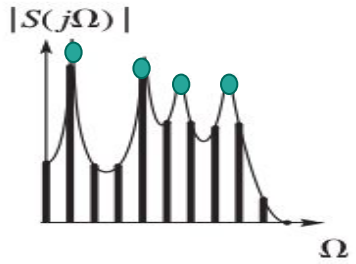
➔ Also known as terminal analogous model

Similar to LTI Systems interacting with input



Marks the gender



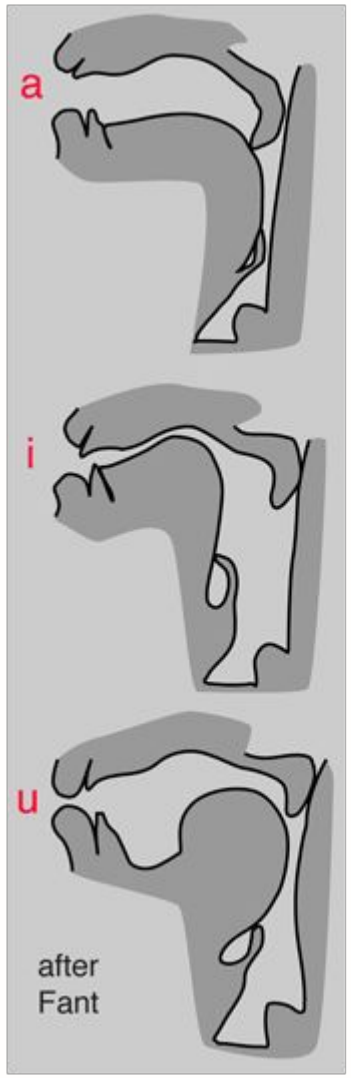


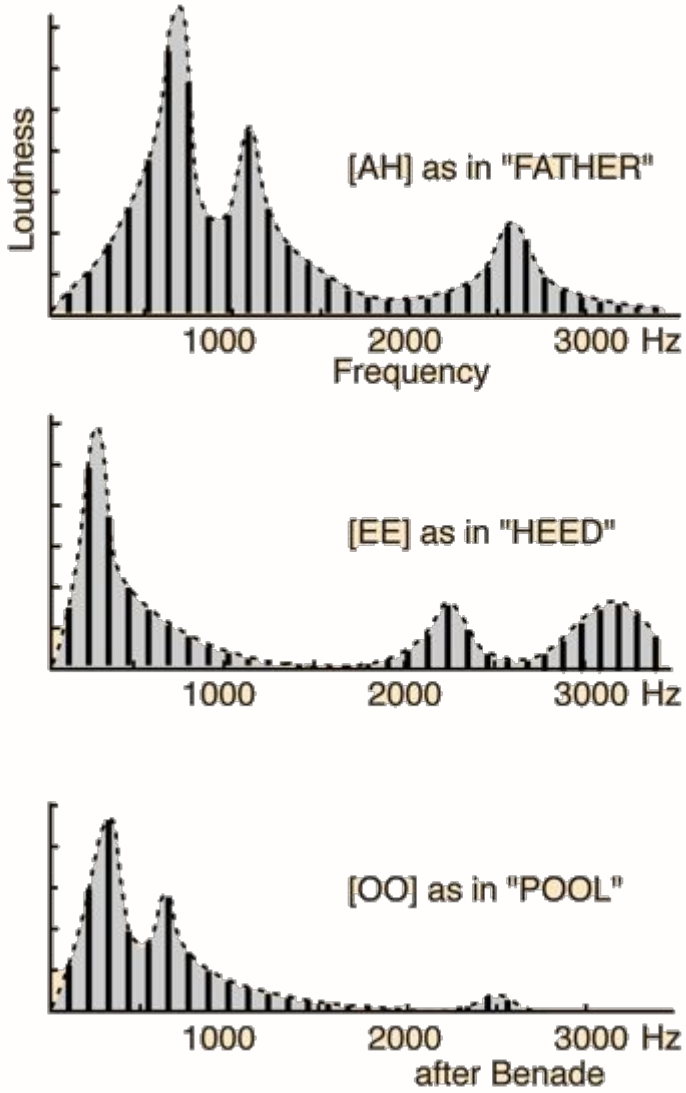
- **Formants** —> Marks the sound

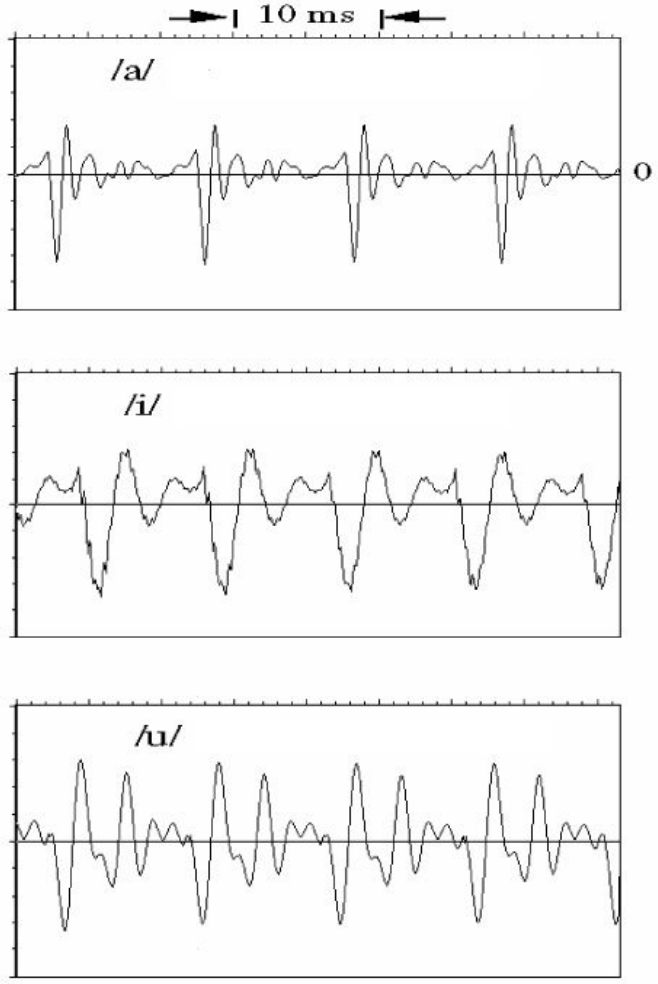
| Vertical lines —> Pitch harmonics

/cs.haifa.ac.il/~nimrod/Compression/Speech/S1Basics2010.pdf

MOUTH CONFIGURATIONS & SOME VOWELS







http://hyperphysics.phy-astr.gsu.edu/hbase/Music/vowel.html ,
http://research.spa.aalto.fi/publications/theses/lemmetty_mst/chap3.html

SHORT TIME FOURIER TRANSFORM (STFT)

Window function



- ➔ **Frequency content changes with time based on spoken words**
- ➔ **Hence STFT is needed**

Multiply x



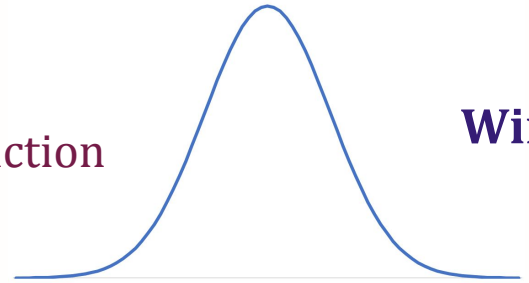


Take Fourier Transform



SHORT TIME FOURIER TRANSFORM (STFT)

Window function



Window moves with some overlap

Multiply

x



Take Fourier Transform

Ref: Theory and Applications of Digital Speech Processing, 1st edition

SHORT TIME FOURIER TRANSFORM (STFT)

Window function



Window moves with some overlap

Multiply x





Log magnitude of the resultant - Spectrogram

Take Fourier Transform



Ref: Theory and Applications of Digital Speech Processing, 1st edition

SPECTROGRAM AND WINDOW LENGTH

Wideband Spectrogram

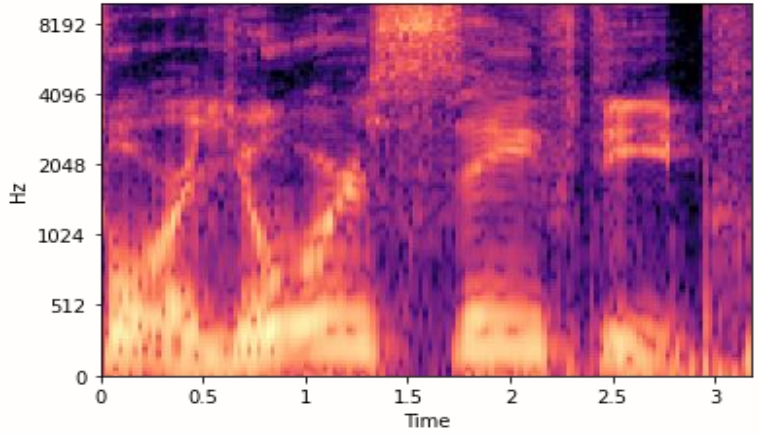
- ➔ Small Time Window
- ➔ Good Time Resolution
- ➔ Captures Formants -
  - ◆ Resonant Frequencies of the vocal tract
  - ◆ Envelope of the spectrum

Narrowband Spectrogram

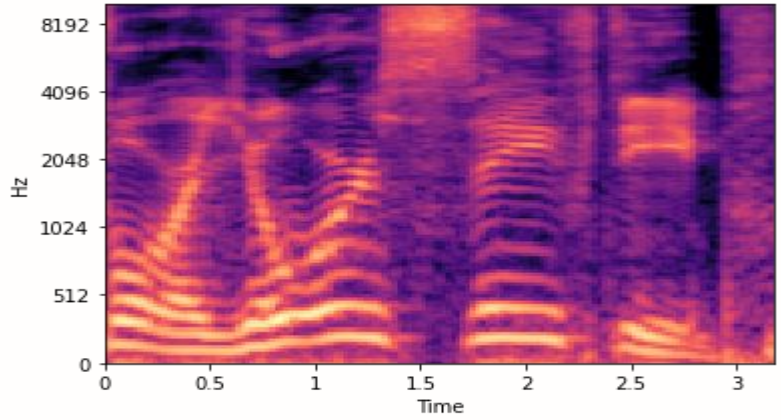
- ➔ Larger Time Window
- ➔ Poor Time Resolution
- ➔ Good Frequency Resolution - Pitch harmonics

SPECTROGRAM & WINDOW LENGTH

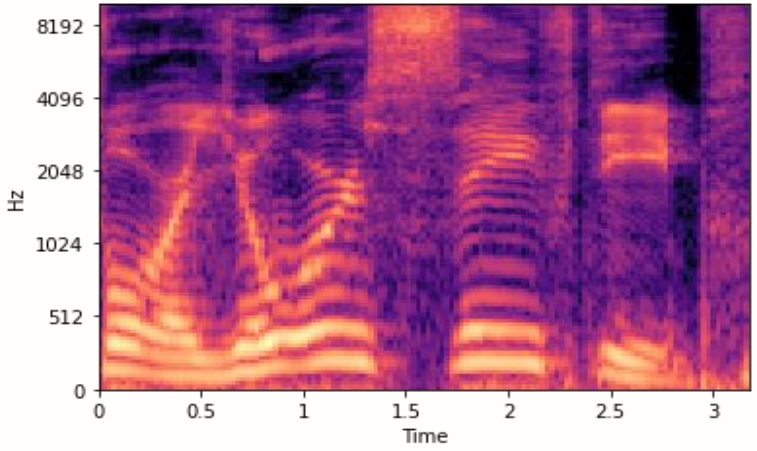
Spectrogram Window - 10 ms



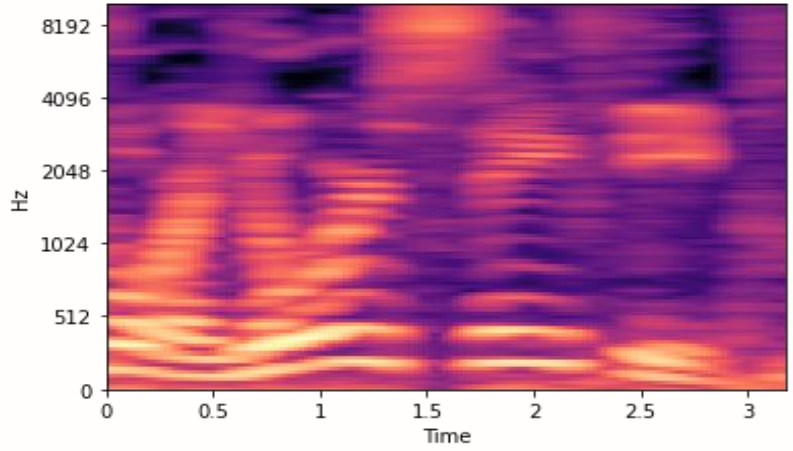
Spectrogram Window - 50 ms



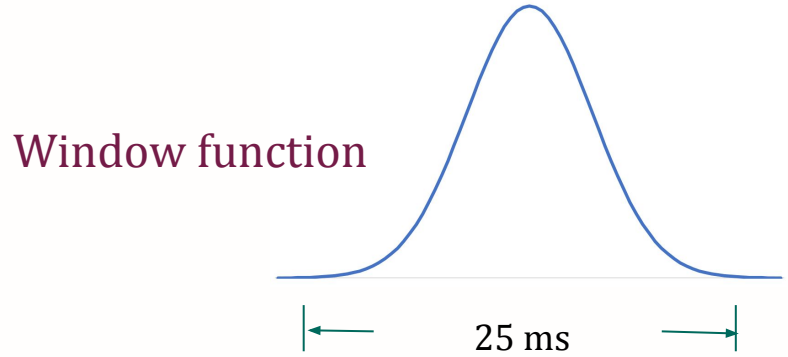
Spectrogram Window - 25 ms



Spectrogram Window - 200 ms



SHORT TIME FOURIER TRANSFORM & SPEECH



In Speech,

➔ Window length = 25 ms

➔ Overlap =10 ms

Multiply X





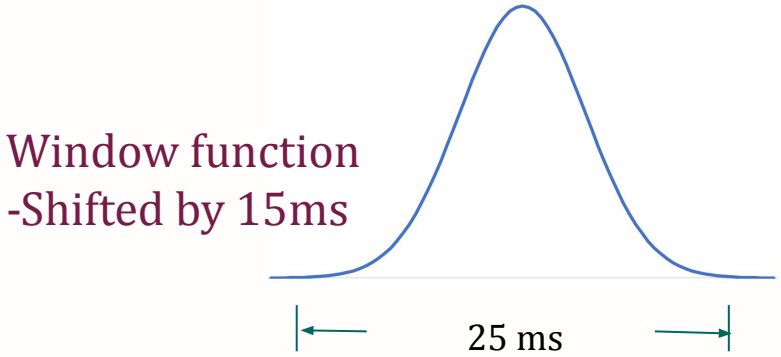
Take Fourier Transform

Log magnitude of the resultant - Spectrogram



Ref: Theory and Applications of Digital Speech Processing, 1st edition

SHORT TIME FOURIER TRANSFORM & SPEECH



In Speech,

➔ Window length = 25 ms

➔ Overlap =10 ms

Multiply X





Take Fourier Transform

Log magnitude of the resultant - Spectrogram

Ref: Theory and Applications of Digital Speech Processing, 1st edition

SPEECH PERCEPTION

- Actual & Perceived Speech Attributes
- Mel Scale
- Masking



SPEECH PERCEPTION



Physical Attribute of Acoustic Signal

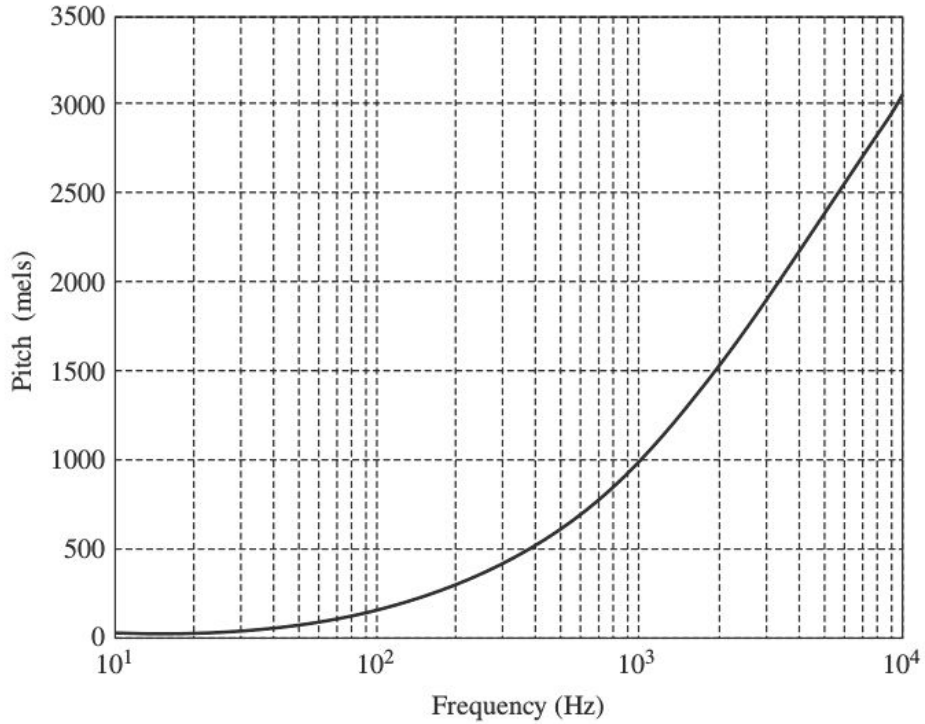
- ➔ Intensity
- ➔ Frequency

Psychophysical Observations

- ➔ Loudness
- ➔ Pitch

PITCH & FREQUENCY

- ➔ It is the perceptual counterpart of frequency
- ➔ Obtained by the frequency perceived by the listener as the signal frequency is varied.
- ➔ Its unit is mel
- ➔ Pitch of a 1000 Hz tone is taken as 1000 mels



Ref: Theory and Applications of Digital Speech Processing, 1st edition

PITCH & LOUDNESS



100Hz



1000Hz



5000Hz



100Hz,
Amp=0.5

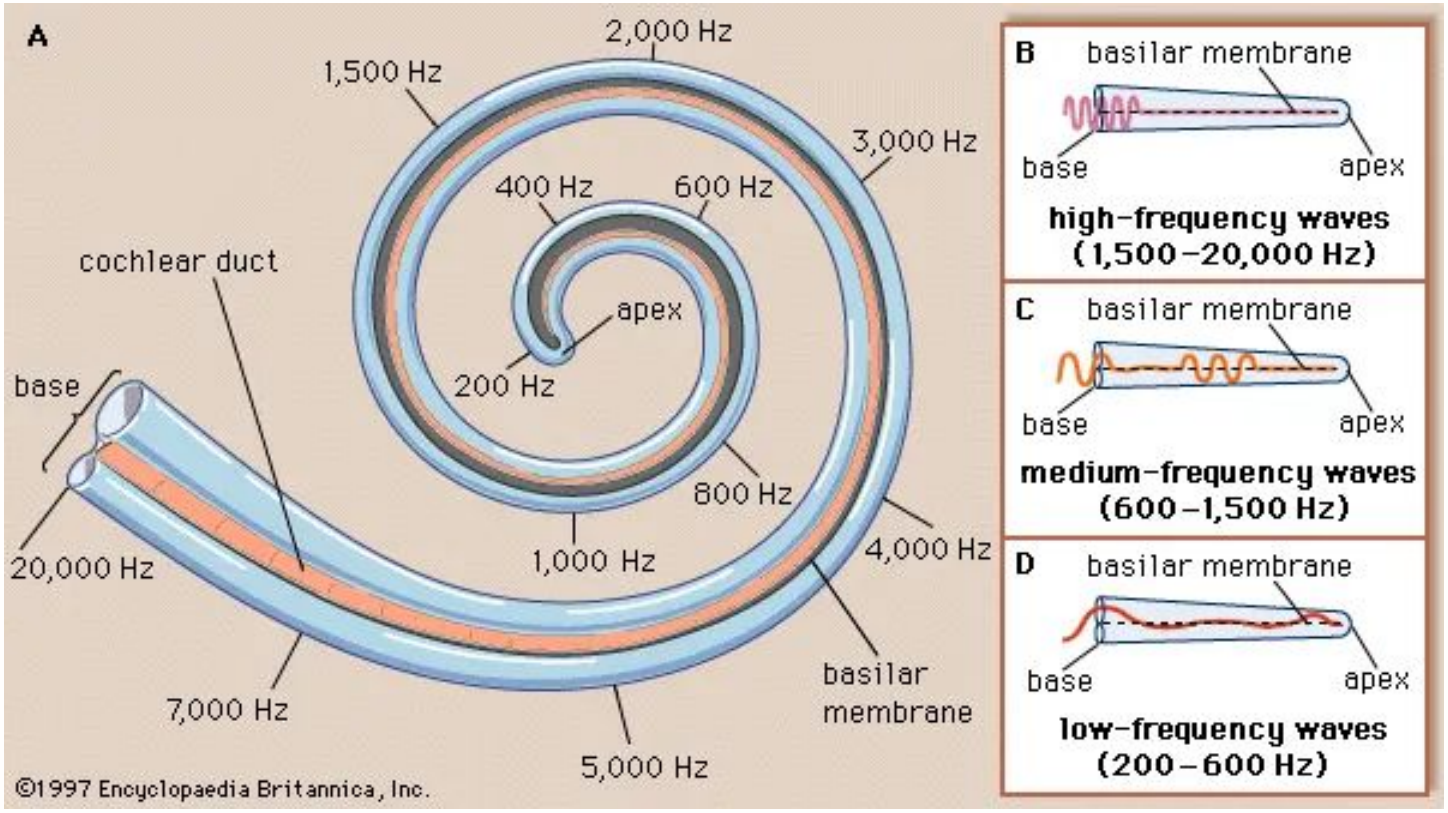


1000Hz
Amp=0.5



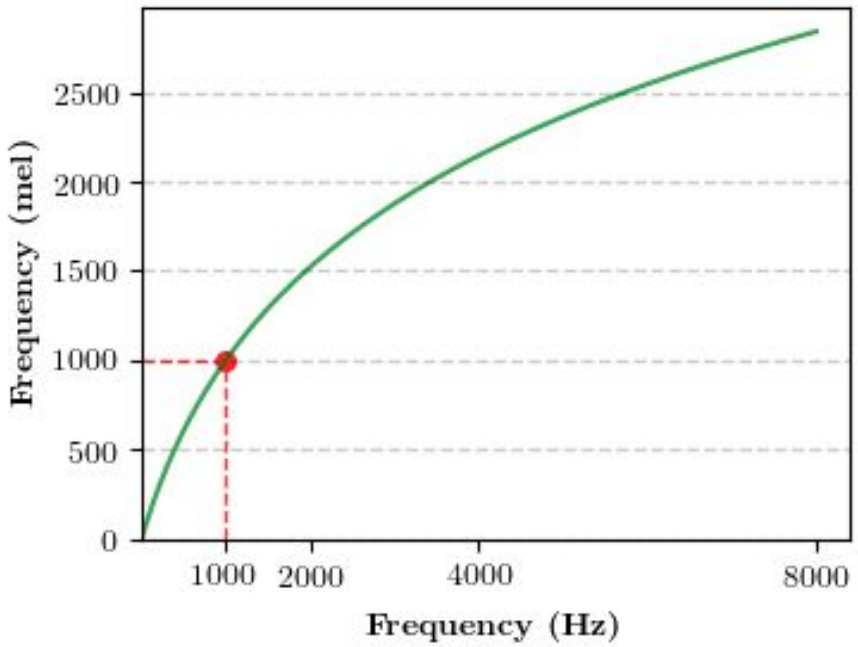
5000Hz
Amp=0.5

BASILAR MEMBRANE - A Frequency Analyser



MEL SCALE

- The mel scale is a quasi-logarithmic function of acoustic frequency designed such that **perceptually similar pitch intervals (e.g. octaves) appear equal in width over the full hearing range.**
- A very important concept in feature extraction of speech signal

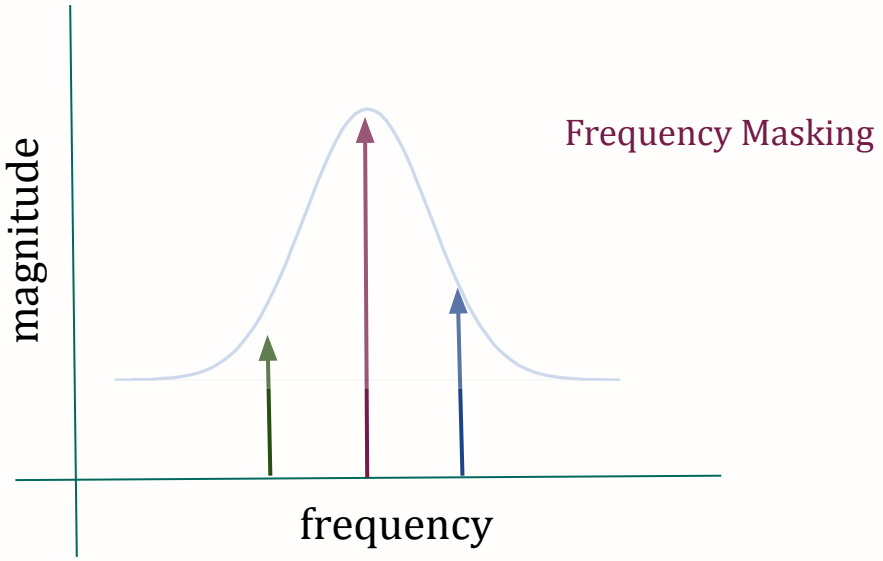


$$\text{Mel} = 1127 \log_e (1 + F/700)$$

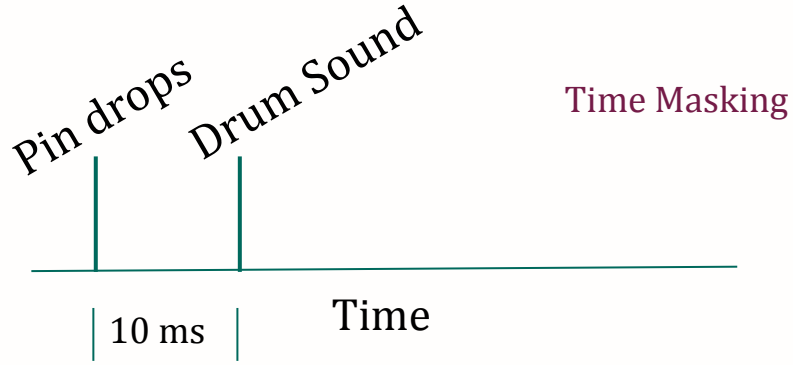
https://www.researchgate.net/figure/The-relationship-between-the-frequency-scale-and-mel-scale-is-shown-on-the-left-The_fig1_336444658

MASKING

- Masking of sounds in presence of other sounds
- A tone can mask other tone
- Noise gets masked too



Frequency Masking



Time Masking

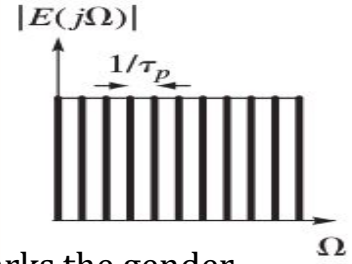
CEPSTRUM



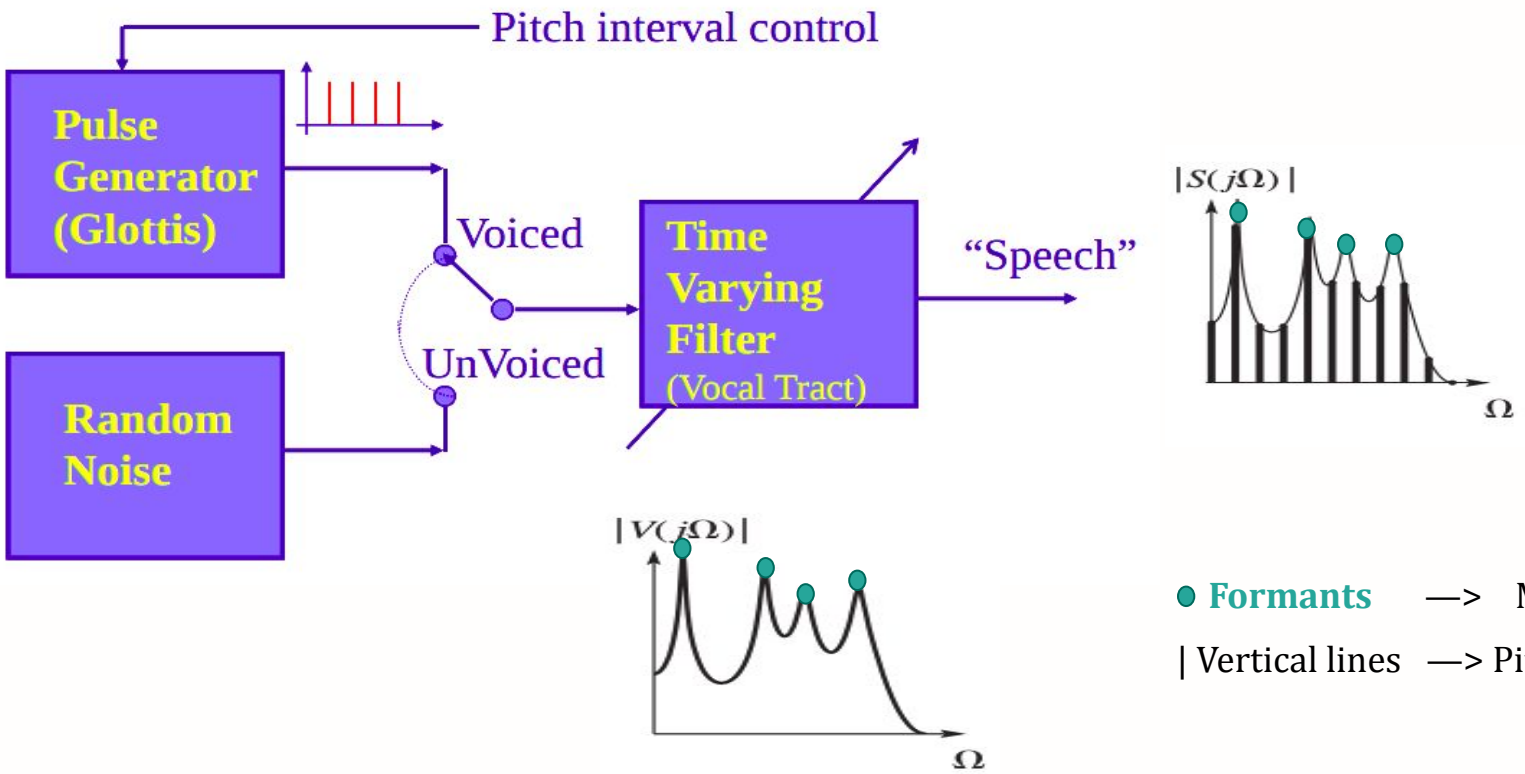
SPEECH PRODUCTION MODEL

➔ It is mathematical equivalent of speech

➔ Also known as terminal analogous model



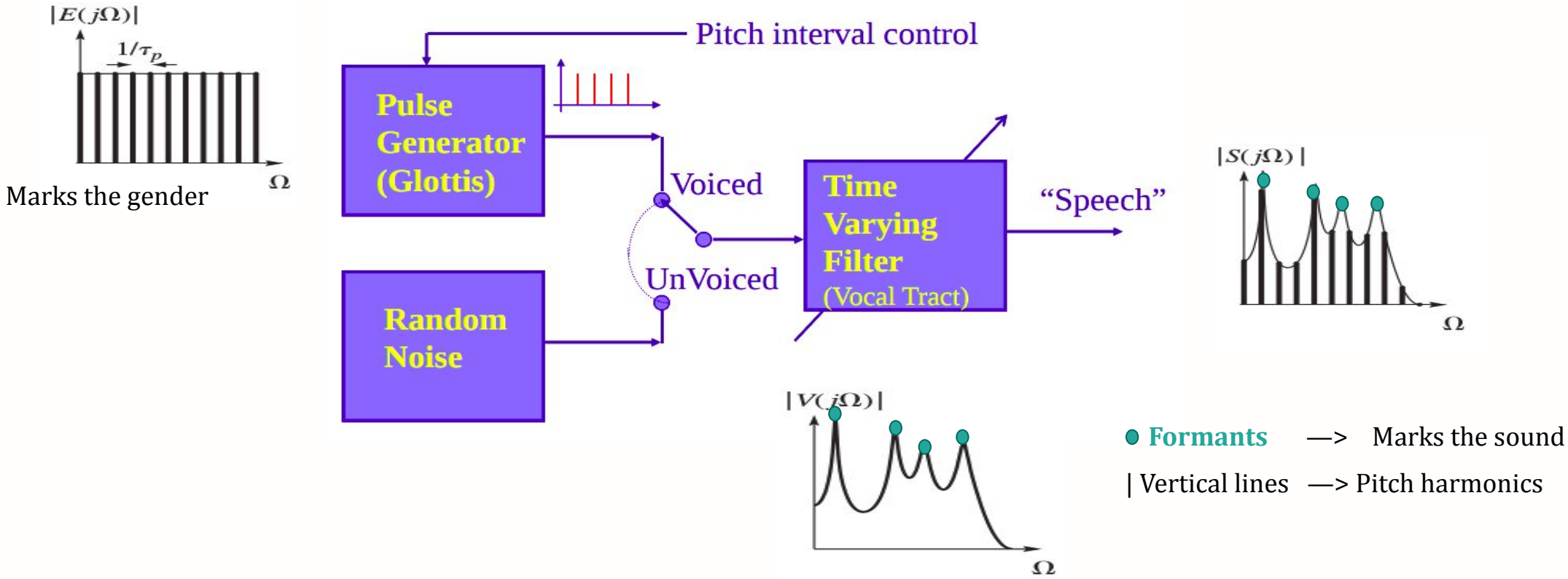
Marks the gender



- **Formants** —> Marks the sound
- | Vertical lines —> Pitch harmonics

http://cs.haifa.ac.il/~nimrod/Compression/Speech/S1Basics2010.pdf

CONVOLUTION IN SPEECH PRODUCTION MODEL



Speech is a result of **convolution** of vocal tract response with glottis pulses/turbulent air

http://cs.haifa.ac.il/~nimrod/Compression/Speech/S1Basics2010.pdf

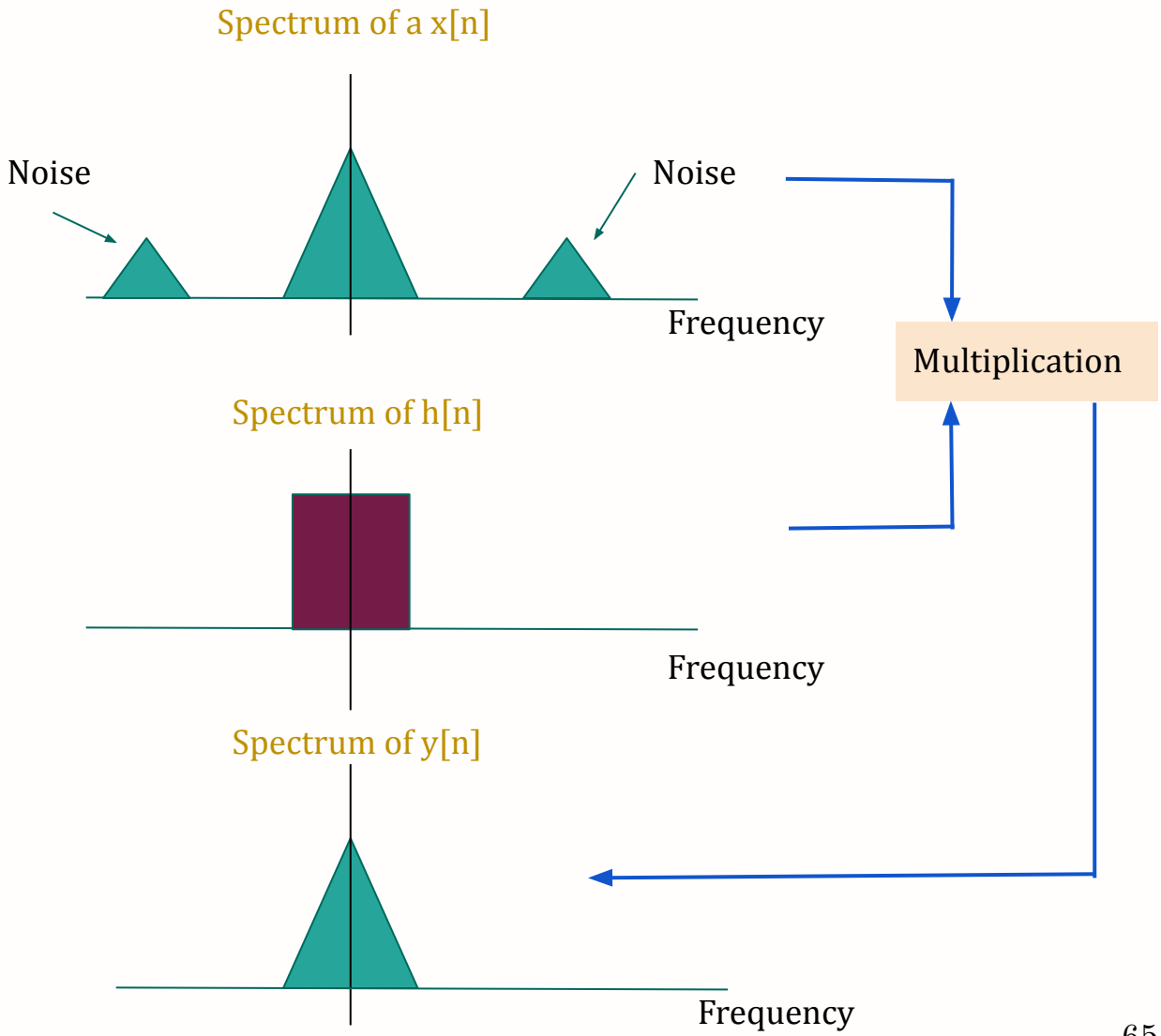
SEPARATION OF VOCAL TRACT RESPONSE

- Objective: We have to separate the vocal tract response
  - Looks like a easy task? - **No**
  - Just do filtering? - **No**
  - When can one do filtering to extract vocal tract response? - **If two signals were added**
  - **What to do if two signals were convolved?**



SEPARATING TWO "ADDED" SIGNALS BY FILTERING

When two signals that are ADDED are occupying different frequency bands we can separate them out by filtering



SEPARATING TWO “CONVOLVED” SIGNALS BY LIFTERING

- ➔ Convolution is **MULTIPLICATION** in frequency-domain.
- ➔ Applying log in frequency domain makes multiplication “**ADDITION**” .
- ➔ Slowly varying vocal-response can then be separated from rapidly varying pitch harmonics! (but since we do it in Frequency domain – called **LIFTERING** (filt-ering is modified as lift-ering)

SEPARATION OF VOCAL TRACT RESPONSE

- **Hint:** Convolution in time domain is multiplication in frequency domain
  $S(j\Omega) = E(j\Omega)*V(j\Omega)$ — **Spectrum domain**

- Which operation converts multiplication to addition? - **Log**
  [log(a*b) = log(a) + log(b)]

  $\log(S(j\Omega)) = \log(E(j\Omega)) + \log(V(j\Omega))$ — **Cepstrum domain**

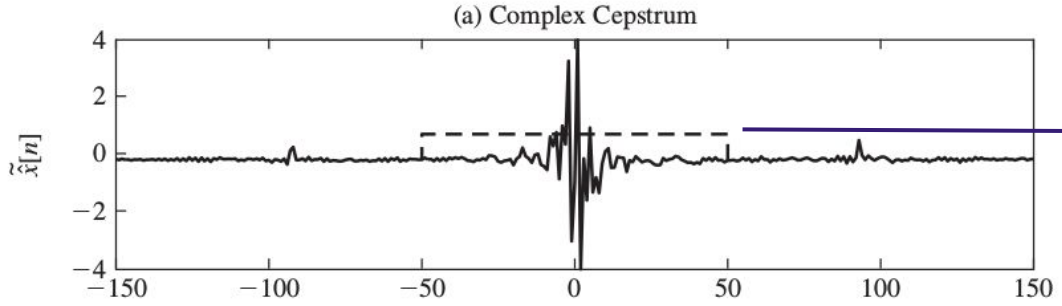
- Which is slowly varying, $\log(E(j\Omega))$ or $\log(V(j\Omega))$? - **$\log(V(j\Omega))$**

- Now, low pass liftering can extract the spectrum related to vocal tract response alone

- A simple low pass lifter - **Rectangular Moving average Lifter**

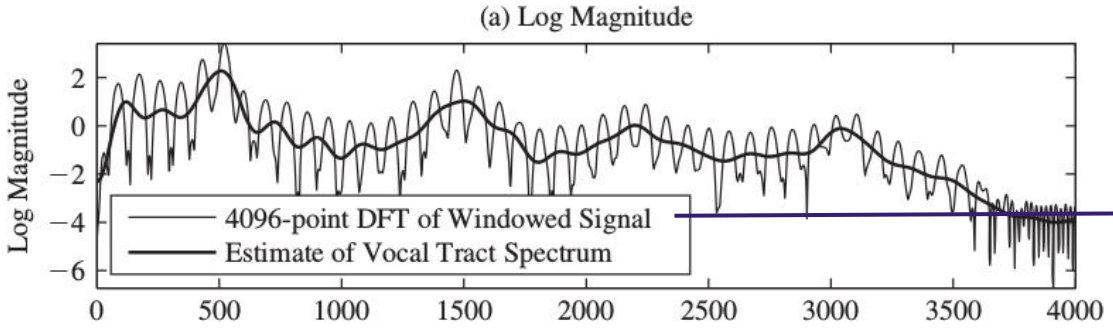


SEPARATION OF VOCAL TRACT RESPONSE



Lifter -

Low quefrency region corresponds to the vocal tract response

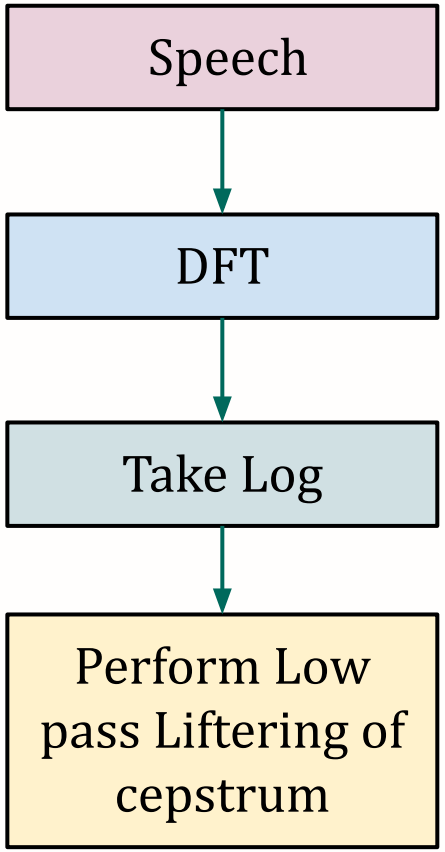


DFT of the windowed signal is the multiplication of excitation spectrum and vocal tract spectrum



Ref: Theory and Applications of Digital Speech Processing, 1st edition

SEPARATION OF VOCAL TRACT RESPONSE



Convolution of excitation impulse/random noise with vocal tract response

Multiplication of excitation impulse/random noise spectrum with vocal tract spectrum

Sum of log magnitude excitation impulse/random noise spectrum with log magnitude vocal tract spectrum

Because vocal tract spectrum is slowly varying
Resultant: Log magnitude spectrum of vocal tract

FEATURES

- Filter bank
- MFCC

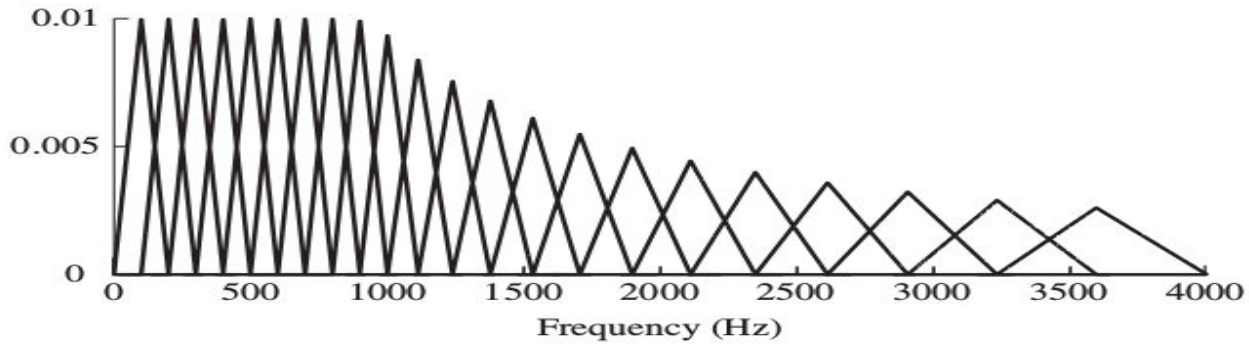


MEL FILTER BANK & MFCCs

➔ In obtaining Mel Filter Bank/MFCC s - **Triangular Moving Average Filters are used on DFT**



➔ The triangular filters are not of the same width and gain - inspired by mel scale (derived from human perception)



Ref: Theory and Applications of Digital Speech Processing, 1st edition

FILTER BANK & MFCCs

- The obtained energy in the filters are named as **filter bank features**
- The obtained filter bank features are correlated (due to overlap), so **DCT-2** is performed **to decorrelate the features to obtain MFCC.**
- This decorrelation helps in assuming diagonal covariance matrices in GMM based models
- Alongwith MFCC, delta and delta features are also taken sometimes
- Δ $MFCC_m[n] = MFCC_m[n] - MFCC_{m-1}[n]$ -> **its too noisy and some sophisticated formula is used in calculating deltas**
- $\Delta\Delta$ $MFCC_m[n]$ = Apply the previous formula on the delta features.
- Usually 13 MFCCs + 13 MFCC deltas + 13 MFCC delta deltas are taken

MEL FILTER BANK & MFCCs

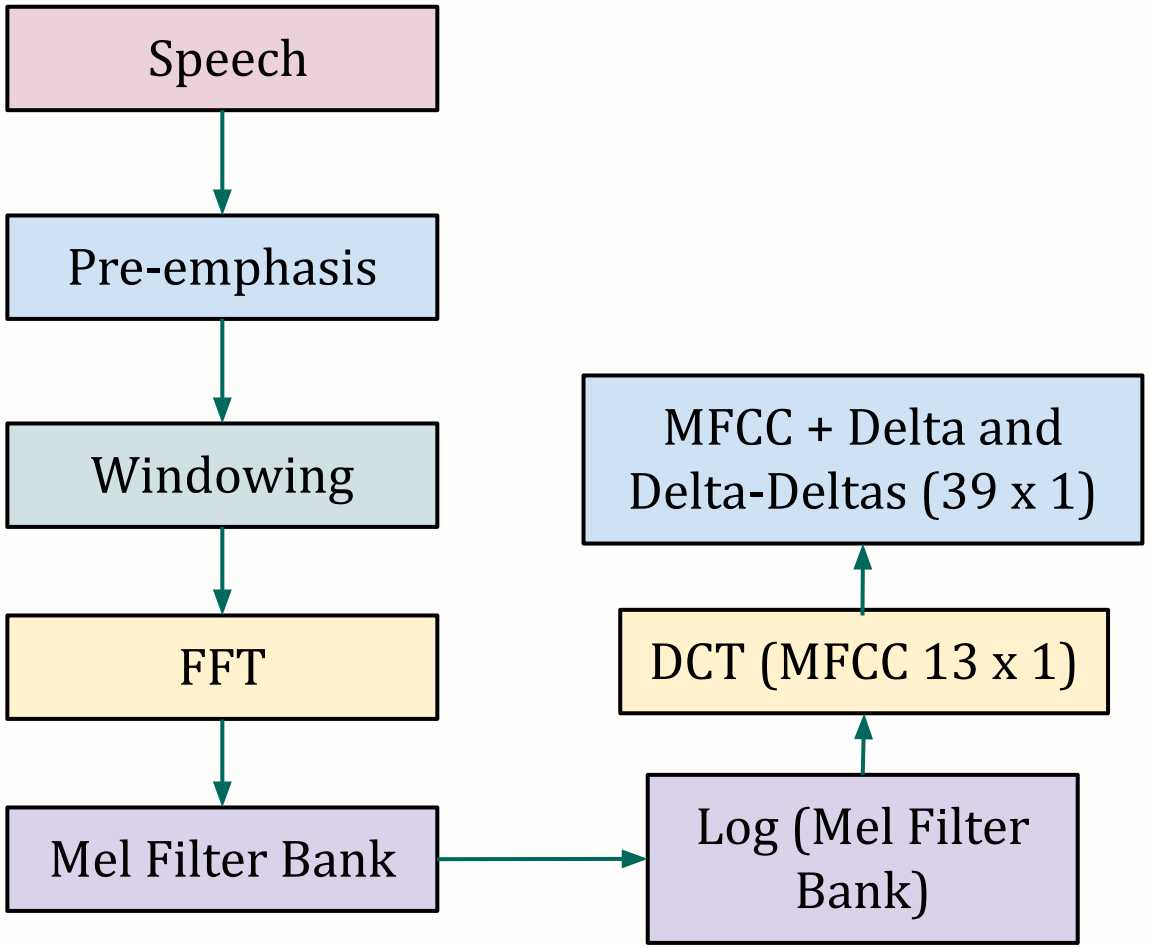
Convolution of excitation impulse/random noise with vocal tract response

All formants to have same magnitude

25 ms window with 10 ms overlap

A computationally effective way of obtaining DFT

Apply Mel Filter bank on DFT and calculate the energy in each filter



DCT 2 for decorrelating the features

Take log to reduce the dynamic range (mimicking humans again - **Cepstrum domain**)